FRM PART I – MOCK II ANSWERS

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By AnalystPrep

Reference Table: Let Z be a standard normal random variable.

z	P(Z<z)	z	P(Z<z)	z	P(Z<z)	z	P(Z<z)	z	P(Z<z)	z	P(Z<z)
-3	0.0013	-2.50	0.0062	-2.00	0.0228	-1.50	0.0668	-1.00	0.1587	-0.50	0.3085
-2.99	0.0014	-2.49	0.0064	-1.99	0.0233	-1.49	0.0681	-0.99	0.1611	-0.49	0.3121
-2.98	0.0014	-2.48	0.0066	-1.98	0.0239	-1.48	0.0694	-0.98	0.1635	-0.48	0.3156
-2.97	0.0015	-2.47	0.0068	-1.97	0.0244	-1.47	0.0708	-0.97	0.1660	-0.47	0.3192
-2.96	0.0015	-2.46	0.0069	-1.96	0.0250	-1.46	0.0721	-0.96	0.1685	-0.46	0.3228
-2.95	0.0016	-2.45	0.0071	-1.95	0.0256	-1.45	0.0735	-0.95	0.1711	-0.45	0.3264
-2.94	0.0016	-2.44	0.0073	-1.94	0.0262	-1.44	0.0749	-0.94	0.1736	-0.44	0.3300
-2.93	0.0017	-2.43	0.0075	-1.93	0.0268	-1.43	0.0764	-0.93	0.1762	-0.43	0.3336
-2.92	0.0018	-2.42	0.0078	-1.92	0.0274	-1.42	0.0778	-0.92	0.1788	-0.42	0.3372
-2.91	0.0018	-2.41	0.0080	-1.91	0.0281	-1.41	0.0793	-0.91	0.1814	-0.41	0.3409
-2.9	0.0019	-2.40	0.0082	-1.90	0.0287	-1.40	0.0808	-0.90	0.1841	-0.40	0.3446
-2.89	0.0019	-2.39	0.0084	-1.89	0.0294	-1.39	0.0823	-0.89	0.1867	-0.39	0.3483
-2.88	0.0020	-2.38	0.0087	-1.88	0.0301	-1.38	0.0838	-0.88	0.1894	-0.38	0.3520
-2.87	0.0021	-2.37	0.0089	-1.87	0.0307	-1.37	0.0853	-0.87	0.1922	-0.37	0.3557
-2.86	0.0021	-2.36	0.0091	-1.86	0.0314	-1.36	0.0869	-0.86	0.1949	-0.36	0.3594
-2.85	0.0022	-2.35	0.0094	-1.85	0.0322	-1.35	0.0885	-0.85	0.1977	-0.35	0.3632
-2.84	0.0023	-2.34	0.0096	-1.84	0.0329	-1.34	0.0901	-0.84	0.2005	-0.34	0.3669
-2.83	0.0023	-2.33	0.0099	-1.83	0.0336	-1.33	0.0918	-0.83	0.2033	-0.33	0.3707
-2.82	0.0024	-2.32	0.0102	-1.82	0.0344	-1.32	0.0934	-0.82	0.2061	-0.32	0.3745
-2.81	0.0025	-2.31	0.0104	-1.81	0.0351	-1.31	0.0951	-0.81	0.2090	-0.31	0.3783
-2.8	0.0026	-2.30	0.0107	-1.80	0.0359	-1.30	0.0968	-0.80	0.2119	-0.30	0.3821
-2.79	0.0026	-2.29	0.0110	-1.79	0.0367	-1.29	0.0985	-0.79	0.2148	-0.29	0.3859
-2.78	0.0027	-2.28	0.0113	-1.78	0.0375	-1.28	0.1003	-0.78	0.2177	-0.28	0.3897
-2.77	0.0028	-2.27	0.0116	-1.77	0.0384	-1.27	0.1020	-0.77	0.2206	-0.27	0.3936
-2.76	0.0029	-2.26	0.0119	-1.76	0.0392	-1.26	0.1038	-0.76	0.2236	-0.26	0.3974
-2.75	0.0030	-2.25	0.0122	-1.75	0.0401	-1.25	0.1056	-0.75	0.2266	-0.25	0.4013
-2.74	0.0031	-2.24	0.0125	-1.74	0.0409	-1.24	0.1075	-0.74	0.2296	-0.24	0.4052
-2.73	0.0032	-2.23	0.0129	-1.73	0.0418	-1.23	0.1093	-0.73	0.2327	-0.23	0.4090
-2.72	0.0033	-2.22	0.0132	-1.72	0.0427	-1.22	0.1112	-0.72	0.2358	-0.22	0.4129
-2.71	0.0034	-2.21	0.0136	-1.71	0.0436	-1.21	0.1131	-0.71	0.2389	-0.21	0.4168
-2.7	0.0035	-2.20	0.0139	-1.70	0.0446	-1.20	0.1151	-0.70	0.2420	-0.20	0.4207
-2.69	0.0036	-2.19	0.0143	-1.69	0.0455	-1.19	0.1170	-0.69	0.2451	-0.19	0.4247
-2.68	0.0037	-2.18	0.0146	-1.68	0.0465	-1.18	0.1190	-0.68	0.2483	-0.18	0.4286
-2.67	0.0038	-2.17	0.0150	-1.67	0.0475	-1.17	0.1210	-0.67	0.2514	-0.17	0.4325
-2.66	0.0039	-2.16	0.0154	-1.66	0.0485	-1.16	0.1230	-0.66	0.2546	-0.16	0.4364
-2.65	0.0040	-2.15	0.0158	-1.65	0.0495	-1.15	0.1251	-0.65	0.2578	-0.15	0.4404
-2.64	0.0041	-2.14	0.0162	-1.64	0.0505	-1.14	0.1271	-0.64	0.2611	-0.14	0.4443
-2.63	0.0043	-2.13	0.0166	-1.63	0.0516	-1.13	0.1292	-0.63	0.2643	-0.13	0.4483
-2.62	0.0044	-2.12	0.0170	-1.62	0.0526	-1.12	0.1314	-0.62	0.2676	-0.12	0.4522
-2.61	0.0045	-2.11	0.0174	-1.61	0.0537	-1.11	0.1335	-0.61	0.2709	-0.11	0.4562
-2.6	0.0047	-2.10	0.0179	-1.60	0.0548	-1.10	0.1357	-0.60	0.2743	-0.10	0.4602
-2.59	0.0048	-2.09	0.0183	-1.59	0.0559	-1.09	0.1379	-0.59	0.2776	-0.09	0.4641
-2.58	0.0049	-2.08	0.0188	-1.58	0.0571	-1.08	0.1401	-0.58	0.2810	-0.08	0.4681
-2.57	0.0051	-2.07	0.0192	-1.57	0.0582	-1.07	0.1423	-0.57	0.2843	-0.07	0.4721
-2.56	0.0052	-2.06	0.0197	-1.56	0.0594	-1.06	0.1446	-0.56	0.2877	-0.06	0.4761
-2.55	0.0054	-2.05	0.0202	-1.55	0.0606	-1.05	0.1469	-0.55	0.2912	-0.05	0.4801
-2.54	0.0055	-2.04	0.0207	-1.54	0.0618	-1.04	0.1492	-0.54	0.2946	-0.04	0.4840
-2.53	0.0057	-2.03	0.0212	-1.53	0.0630	-1.03	0.1515	-0.53	0.2981	-0.03	0.4880
-2.52	0.0059	-2.02	0.0217	-1.52	0.0643	-1.02	0.1539	-0.52	0.3015	-0.02	0.4920
-2.51	0.0060	-2.01	0.0222	-1.51	0.0655	-1.01	0.1562	-0.51	0.3050	-0.01	0.4960

Reference Table: t-table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

Foundations of Risk Management

1. Next week, Global Tech has declared its intention to bring to the market a 15-year senior bond issue at par with a coupon rate of 18%, offering a spread of 800 basis points over the corresponding 15-year Treasury issue. An investor is keen to enter into a total return swap that matures in one year with the senior bonds that are about to be issued as the reference obligation. Under the terms of the contract, payments will be exchanged semiannually, where the total return receiver will pay the six-month Treasury rate plus 350 basis points.

What is the 15-year Treasury rate at the time the bonds are issued?

A. 8%
B. 2%
C. 10%
D. 4.5

Correct answer: C

Explanation: We know that Global Tech will be coming to market with a 15-year senior bond issue at par with a coupon rate of 18%, offering a spread of 800 basis points over the 15-year Treasury issue. Because 800 basis points is 8%, this means the 15-year Treasury issue will have a rate of 18% - 8% = 10%. At the time of issuance, therefore, the 15-year Treasury yield is 10%. Note that the question has a bit of nugatory information.

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 4, Credit Risk Transfer Mechanisms, Learning objective: Compare different types of credit derivatives, explain how each one transfers credit risk, and describe their advantages and disadvantages.

2. A portfolio has a return of 15%, and its standard deviation is 25%. If the return on the market is 8% with a standard deviation of 50% and the risk-free rate is 5%, what are the Sharpe and Treynor ratios for the portfolio?

A. Sharpe ratio: 0.40; Treynor ratio: 0.13
B. Sharpe ratio: 0.28; Treynor ratio: 0.09
C. Sharpe ratio: 0.40; Treynor ratio: 0.20
D. Sharpe ratio: 0.28; Treynor ratio: 0.09

Correct answer: C

Explanation: Sharpe Ratio = $\frac{R_p - R_f}{\sigma}$

$$= \frac{0.15 - 0.05}{0.25} = 0.4$$

Treynor Ratio = $\frac{Rp - Rf}{\beta}$

$$\beta = \frac{\sigma_p}{\sigma_m} = \frac{0.25}{0.5} = 0.5$$

$$= \frac{0.15 - 0.05}{0.5} = 0.20$$

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 5: Modern Portfolio Theory (MPT) and the Capital Asset Pricing Model (CAPM), Learning objective: Calculate, compare, and interpret the following performance measures: the Sharpe performance index, the Treynor performance index, the Jensen performance index, the tracking error, information ratio, and Sortino ratio.

3. You have been given the following information regarding a portfolio:

Return on the portfolio	12%
Return on the market	8%
Return on benchmark	9%
Beta of the portfolio	1.1
Tracking error of the portfolio	10%
Standard deviation of the portfolio	15%

Which of the following is closest to the information ratio of the portfolio?

A. 0.3
B. 0.2
C. 0.4
D. 0.5

Correct answer: A

Explanation: Information Ratio = $\frac{Rp - RB}{Tracking\ error}$

$$= \frac{12\% - 9\%}{10\%} = 0.30$$

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 5: Modern Portfolio Theory (MPT) and the Capital Asset Pricing Model (CAPM), Learning objective: Calculate, compare, and interpret the following performance measures: the Sharpe performance index, the Treynor performance index, the Jensen performance index, the tracking error, information ratio, and Sortino ratio.

4. The collapse of Enron was most likely a result of:

A. Unauthorized trading activity
B. Unanticipated market moves
C. Dubious accounting practices
D. All of the above

Correct answer: C

Explanation*:* Enron was involved in dubious accounting practices. Under pressure from shareholders, company executives began to rely on manipulative accounting practices, including a technique known as "mark-to-market accounting," to hide the troubles. Mark-to-market accounting allowed the company to write unrealized future gains from some trading contracts into current income statements, thus giving the illusion of higher current profits.

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 9: Learning From Financial Disasters, Learning objective: Corporate governance, including the Enron case

5. Which of the following cases most likely involved the mismanagement of hedging strategies?

A. Enron
B. Volkswagen
C. Metallgesellschaft
D. Barings Bank

Correct answer: C

Explanation: The financial failure of Metallgesellschaft was a result of the company's hedging of long-term oil and gas contracts with short-term oil and gas futures. The futures required cash settlement and a large adverse movement in the price of oil and gas caused large funding liquidity problems for the company.

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 9: Learning From Financial Disasters, Learning objective: Implementing hedging strategies, including the Metallgesellschaft case.

6. The 2007-2008 financial crisis was primarily a result of two major risks. These are:

A. Credit and liquidity risk
B. Credit and market risk
C. Liquidity and operational risk
D. Market and operational risk

Correct answer: A

Explanation: The 2007-2008 financial crisis was primarily caused by the mismanagement of credit risks, which also resulted in a liquidity crunch. During a liquidity crunch, businesses and consumers are charged high-interest rates on loans, which are more difficult to obtain.

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 10: Anatomy of the Great Financial Crisis of 2007-2009

Learning objectives:

- ***Describe the historical background and provide an overview of the 2007-2009 financial crisis.***
- ***Describe the build-up to the financial crisis and the factors that played an important role.***

7. A bank only uses VaR to measure the risk in its trading book. The bank has a one-day VaR of $25 million at a confidence level of 99%. During the last 200 trading days, this VaR was exceeded only twice, with total losses of $500 million and $200 million being incurred. According to the VaR measure on its own, the bank has been successful in managing its risk but, in reality, it has lost $700 million.

In this scenario, the failure of risk management can *most likely* be attributed to:

A. Failure to use appropriate risk metrics
B. Mismeasurement of known risk
C. Failure to take known risks into account
D. Failure in communicating risk to top management

Correct answer: A

Explanation: There are five different types of risk management failures. These arise from:

- Failure to use appropriate risk measures
- Mismeasurement of known risks
- Failure to take known risks into account
- Failure in communicating known risks to top management
- Failure in monitoring and measurement of risks

In this case, the failure arises because the bank has failed to use an appropriate risk measure. VaR alone does not cater for catastrophic risks but only monitors the frequency of losses above a certain level regardless of their absolute value.

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 7. Risk Data Aggregation and Reporting Principles, Learning objective: Describe key governance principles related to risk data aggregation and risk reporting practices

8. Which of these correctly describes the numerator of the Sortino measure?

A. Difference between portfolio return and market return
B. Difference between portfolio return and minimum acceptable return
C. Difference between portfolio return and risk-free rate
D. Difference between portfolio return and benchmark return

Correct answer: B

Explanation: The Sortino ratio measures the ratio of the difference between portfolio returns and minimum acceptable returns and the semivariance (which measures the variability of the portfolio's returns below minimum acceptable return). Unlike most other ratios such as the Sharpe ratio, the Sortino ratio uses the downside deviation as its risk measure, thereby addressing the problem of using total risk (because upside risk is beneficial to investors.

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 5: Modern Portfolio Theory (MPT) and the Capital Asset Pricing Model (CAPM), Learning objective: Calculate, compare, and interpret the following performance measures: the Sharpe performance index, the Treynor performance index, the Jensen performance index, the tracking error, information ratio, and Sortino ratio.

9. Which of the following parts of GARP's Code of conduct requires members to comply with all applicable laws, rules, and regulations (including the Code itself) governing the GARP Members' professional activities?

A. Fundamental responsibilities
B. Professional integrity and ethical conduct
C. Conflict of interest
D. General accepted principles

Correct answer: A

Explanation: GARP's Code of conduct related to Fundamental responsibilities states that the member shall:

1. Comply with all applicable laws, rules, and regulations (including this Code) governing the GARP Members' professional activities

2. Not outsource or delegate those responsibilities to others

3. Be diligent about not overstating the accuracy or certainty of results or conclusions

4. Clearly disclose the limits of their expertise and knowledge in areas of risk assessment

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 11: GARP Code of Conduct, Learning objective: Describe the responsibility of each GARP Member with respect to professional

integrity, ethical conduct, conflicts of interest, the confidentiality of information, and adherence to generally accepted practices in risk management.

10. Jane Johnson, FRM, works as a full-time risk manager for the Washington Investment Banking Group. Recently, one of Johnson's long-time friends asked her to help with preparations for an initial public offering (IPO) for a promising tech company based in Silicon Valley. This would be a part-time role and would entail establishing earnings projections for the tech company. Johnson should most likely:

A. Accept the work as long as the Washington Investment Banking Group agrees to all the terms of engagement
B. Accept the work as long as she obtains written consent from the Washington Investment Banking Group and does it in her free time
C. Accept the role as long as, in her own assessment, it does not interfere with her duties to the Washington Investment Banking Group
D. Not accept the work as it violates the Code of Conduct by creating a conflict of interest

Correct answer: A

Explanation: The Code of Conduct dictates members to make full and fair disclosure of all matters that could either impair independence and objectivity or interfere with respective duties to their employer, clients or potential clients. Johnson should only accept the role if her current employer fully agrees to the terms.

Option B is incorrect. Johnson doesn't have to do it in her free time; she can be paid for the work done for the tech company.

Option C is incorrect. Johnson has to obtain consent to all the terms of the engagement from her current employer.

Option D is incorrect. Establishing earnings projections for the tech company does not interfere with her rose as a risk manager for the Washington Investment Banking Group.

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 11: GARP Code of Conduct, Learning objectives:

- ***Describe the responsibility of each GARP Member with respect to professional integrity, ethical conduct, conflicts of interest, the confidentiality of information, and adherence to generally accepted practices in risk management.***
- ***Describe the potential consequences of violating the GARP Code of Conduct.***

11. Peter Drury, FRM, serves as the chief risk officer at Capital bank. He recently finalized a comprehensive risk assessment on a series of investments in credit default swaps. According to his estimates, the portfolio had a 1% chance of losing USD 500 million or more over one year, a big enough loss to trigger insolvency or attract takeover bids. Based on Drury's assessment, Capital Bank's CEO authorized a large investment in additional swaps. At the end of the first year, the portfolio had lost USD 650 million. The bank was immediately closed down by the regulator, pending a comprehensive audit and review of the bank's investment policies.

Which of the following statements is correct?

A. The outcome demonstrates risk management failure because the CRO failed to foresee a move by the regulator, neither did he attempt to stop the shutdown
B. The outcome demonstrates an outright risk management failure because the occurrence of a supposedly extreme event means the probability of such an outcome was poorly estimated
C. The outcome demonstrates risk management failure because the CRO did not eliminate the chance of a financial loss
D. On the basis if the information provided, one cannot conclusively determine whether this was a risk management failure

Correct answer: D.

Explanation: In practice, it's the CEO's role to decide on all firm investments, not the CRO's – and the risk management unit in extension. The CRO had correctly predicted that there was a chance of losing USD 500 million or more. What's more, we've not been given information on the distribution of VaR. In these circumstances, this could have been a case of bad luck. An outright risk management failure would have occurred if the CRO had given the occurrence of such an outcome a probability of zero.

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 7. Risk Data Aggregation and Reporting Principles, Learning objective: Describe key governance principles related to risk data aggregation and risk reporting practices

12. A risk manager at an investment consultancy firm is projecting a return of 14% on Portfolio Y. The market risk premium is 15%, the risk-free rate on interest is 5%, and the volatility of the market portfolio is 18%. Portfolio Y has a beta of 0.6. According to the Capital Asset Pricing Model, which of the following statements is true?

A. The expected return of portfolio Y is more than the expected return of the market portfolio
B. The return of the portfolio has more volatility than the market portfolio
C. The expected return of portfolio Y is less than the expected return of the market portfolio
D. The expected return of portfolio Y is equal to the expected return of the market portfolio

Correct answer: C.

Explanation: According to the CAPM, the required return on portfolio Y is:

$$R_y = R_f + \beta\left[E(R_m) - R_f\right] = 5 + 0.6(15) = 14\%$$

Since the beta is less than 1, the required return on Y must be less than the expected return on the market, which is 15%. Note that the question has a lot of immaterial information. As such, a good understanding of CAPM is crucial.

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 5: Modern Portfolio Theory (MPT) and the Capital Asset Pricing Model (CAPM), Learning objective: Apply the CAPM in calculating the expected return on an asset.

13. Catherine Austin works at the arbitrage department of an investment bank. She finds that an asset is trading at USD 2,200.The price of a one-year futures contract on the asset is USD2,244, and the price of a two-year futures contract is USD 2,295.50. Assume that the asset exhibits no cash flows in the first two years. If the term structure of interest rates is flat at 2%, which of the following would be an appropriate arbitrage strategy?

A. Short the 2-year futures, borrow USD 2,000 at 2% and buy the underlying asset
B. Buy the 2-year futures and short the underlying asset
C. Short the 2-year futures, borrow USD 2,000 at 2% and short the underlying asset
D. Buy the 2-year futures, borrow USD 2,000 at 2% and short the underlying asset

Correct answer: A

Explanation: The 1-year futures contract price should be $2200 \exp^{0.02} = 2{,}244$

The 2-year futures price should be $2200 \exp^{0.02\times 2} = 2{,}289.78$

As can been seen, the 2-year futures price (2,295.50) is overvalued compared to the theoretical price (2,289.78). To lock in a profit, one could short the 2-year futures, borrow USD 2,200 at 2%, and use the funds to buy the underlying asset. At the end of 2 years, sell the asset at USD 2,295. Then return the borrowed funds plus interest, which would amount to $2{,}200 \exp^{0.02\times 2} = 2{,}289.78$. At the end of the day, there would be a gain of USD 5.72 ($= 2{,}295.50 - 2289.78$)

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 6. Multifactor Models of Risk-Adjusted Asset Returns, Learning objective: Calculate the expected return of an asset using a single-factor and a multifactor model.

14. A wealth management firm is contemplating motivational measures in an attempt to align the interests of managers and shareholders. To this end, the board has plans to grant managers securities that are tied to the firm's stock. Even as it ponders such a move, the board is wary of managerial incentives that could negatively affect risk management at the firm. Which of the following securities provides the best example of rewards likely to reduce managerial incentives to manage risk?

A. A long position in the firm's stock
B. A deep out-of-the-money call option on the firm's stock
C. At-the-money call option on the firm's stock
D. A deep in-the-money call option on the firm's stock

Correct answer: B.

Explanation: Deep in-the-money call options would only have value if the firm experiences a significant upturn in performance, increasing firm value. And because pulling off a major improvement in firm value is normally a mean task, the management could resort to riskier business dealings with high potential for big profits but also high potential for big losses. Managers could even attempt to further improve chances of more than average returns by reducing hedging, e.g, insurance and use of CDSs, and hence saving on associated costs such as premiums. An at-the-money call could bring about similar results, but managers would not have to for as large a stick price increase. A long position in the firm's stock would have an effect that's somewhat similar to that of a deep in-the-money call.

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 6. Multifactor Models of Risk-Adjusted Asset Return, Learning objective: Explain the arbitrage pricing theory (APT), describe its assumptions, and compare the APT to the CAPM.

15. The failure of Long Term Capital Management almost brought the financial markets to a standstill and has since become a marker buoy to show just how far certain risk management safeguards can go to alleviate the risk of failure. Which of the following statements about LTCM's failure is incorrect?

A. LTCM relied too much on theoretical market risk models and too little on stress testing, gap risk, and liquidity risk
B. LTCM scantily disclosed its positions and exposures, and counterparties had few ways to find out LTCM's exposure to other counterparties
C. In most of its repo and swap transactions' LTCM did not pay an initial margin as would other non-bank counterparties
D. There was no moral hazard at play in the run-up to LTCM's failure

Correct answer: D

Explanation: Moral hazard did play a role in LTCM's failure. The readiness of the US Federal Reserve System to bail out troubled financial institutions, including LTCM, increased the latters' appetite for highly risky positions.

There are several culprits and events that collectively played a role in LCTM's failure. At the heart of its poor risk management was LTCM's overreliance on theoretical market risk models at the expense of stress testing, while still turning a blind eye to gap risk and liquidity risk. There was an exaggerated assumption that LTCM's portfolio was sufficiently diversified across markets. However, in most markets, LTCM was replicating basically the same credit spread trade.

In addition, LTCM made little effort to disclose its exposures to counterparties, which in part explains its leverage. There was no way counterparties could establish LTCM's exposure to other counterparties.

Unlike other non-bank participants in swaps and repo transactions, LTCM was not asked to pay initial margins. As a result, LTCM was able to build layer upon layer of repos and swaps.

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 9: Learning From Financial Disasters, Learning objective: Model risk, including the Niederhoffer case, Long Term Capital Management, and the London Whale case

16. A portfolio manager returns 20% with a volatility of 30%. The benchmark returns 15% with a risk of 22%. The correlation between the two is 0.80, while the risk-free rate is 5%. Which of the following statements is correct?

A. The portfolio has a lower SR than the benchmark
B. The IR of the portfolio is negative
C. IR = 0.35
D. IR = 0.28

Correct answer: D

Explanation: Sharpe ratio of portfolio $= \frac{20\%-5\%}{30\%} = 0.50$

Sharpe ratio of benchmark $= \frac{15\%-5\%}{22\%} = 0.45$

Thus, the sharp ratio of the portfolio is higher than that of the benchmark. A is incorrect

$\text{TEV} = \sqrt{30\%^2 + 22\%^2 - 2 \times 0.80 \times 30\% \times 22\%} = \sqrt{0.0328} = 0.1811 = 18.1\%$

Thus, the IR of the portfolio $= \frac{20\%-15\%}{18.1\%} = 0.28$

Thus, the IR is positive. B is incorrect

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 5: Modern Portfolio Theory (MPT) and the Capital Asset Pricing Model (CAPM), Learning objective: Calculate, compare, and interpret the following performance measures: the Sharpe performance index, the Treynor performance index, the Jensen performance index, the tracking error, information ratio, and Sortino ratio.

17. Patricia Rice, FRM, is evaluating the risk management of Bright Technologies. She is asked to categorize the following events into various types of risk. Identify each numbered event as a credit risk, market risk, business risk, operational risk, and legal risk.

1. A delayed launch of the latest model of an internet connectivity gadget leads to an inventory pileup of the older product and a USD 100 million write-off of excess inventory.

2. Insufficient training leads to misuse of the check clearing system

3. Credit spreads widen following bankruptcies

4. Swap seller does not have the resources required to honor a contract

5. Credit swaps cannot be netted because they originated in Canada and Sydney

A. 1: operational risk; 2: business risk; 3: credit risk; 4: market risk; 5: credit risk
B. 1: business risk; 2: operational risk; 3: credit risk; 4: market risk; 5: credit risk
C. 1: business risk; 2: operational risk; 3: market risk; 4: credit risk; 5: legal risk
D. 1: operational risk; 2: business risk; 3: credit risk; 4: market risk; 5: legal risk

Correct answer: C.

Explanation: A delayed product launch and related inventory losses is an example of business risk

Insufficient training that leads to misuse of the check clearing system is an example of operational risk

Wider credit spreads represent an increase in market risk

A swap seller failing to honor contractual obligations is a credit risk event

When a contract is originated in multiple jurisdictions, enforceability problems may prevail, forming part of legal risk.

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 1: The Building Blocks of Risk Management, Learning objective: Describe and differentiate between the key classes of risks, explain how each type of risk can arise, and assess the potential impact of each type of risk on an organization.

18. Which of the following risk management responsibilities rest with the board of directors of a bank?

I. Approval of the bank's risk management policies and procedures
II. Maintaining oversight of all risk management activities
III. Evaluating the performance of risk management activities
IV. Ensuring compliance with minimum or best practice standards

A. I and II only
B. I and III only
C. II and IV only
D. I, II, III, and IV

Correct answer: D

Explanation: It's the board's role to approve policies, evaluate and maintain oversight of risk management, including auditing of risk management practices to ensure compliance with minimum or best practice standards. IV is actually a responsibility of the audit committee of the board.

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 3: The Governance and Risk Management, Learning objective: Assess the role and responsibilities of the board of directors in risk governance

19. Mathew Jelkins, FRM, was recently found guilty of violating the principle of professional integrity and ethical conduct as outlined in the GARP Code of Conduct. Which of the following is a potential consequence? Jelkins will be:

A. Required to participate in at least one ethical training workshop
B. Stripped of the GARP member's right to use the FRM designation
C. Stripped of the GARP member's right to work in the risk management profession
D. Required to pay a fine whose amount will be determined by the GARP's penalty committee

Correct answer: B

Explanation: According to the GARP Code of Conduct, violations of the Code may result in, among other things, the temporary suspension or permanent removal of the GARP Member from GARP's Membership roles and may also include temporarily or permanently removing from the violator the right to use of refer to having earned the FRM designation or any other GARP-granted designation.

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 11: GARP Code of Conduct, Learning objective: Describe the potential consequences of violating the GARP Code of Conduct.

20. You have been given the following data for a managed portfolio:

Beta	1.55
Alpha	2.11%
Average return	12.5%
Risk-free rate	3%

Basing your calculations on Jensen's measure of portfolio performance, the return on the market portfolio is closest to:

A. 1.77%
B. 7.77%
C. 8.36%
D. 5.36%

Correct answer: B

Explanation: According to Jensen's measure of performance:

$\alpha_p = E(R_p) - [R_f + \beta_p[E(R_m) - R_f]]$

$2.11\% = 12.5\% - [3\% + 1.55(x - 3\%)]$

$2.11\% = 12.5\% - [3\% + 1.55x - 4.65\%]$

$2.11\% = 12.5\% - 3\% - 1.55x + 4.65\%$

$1.55x = 12.04\%$

$x = E(R_f) = 7.77\%$

FRM Part 1, Book 1: Foundations of Risk Management, Chapter 5: Modern Portfolio Theory (MPT) and the Capital Asset Pricing Model (CAPM), Learning objective: Calculate, compare, and interpret the following performance measures: the Sharpe performance index, the Treynor performance index, the Jensen performance index, the tracking error, information ratio, and Sortino ratio.

Quantitative Analysis

21. Which of the following statements is false?

A. The result of a single toss of a fair coin is a mutually exclusive event
B. The results of two coin tosses of a fair coin are independent events
C. The results of drawing two cards from a deck of cards are independent events
D. The results of two rolls of a fair die are independent events

Correct answer: C

Explanation: Two events are mutually exclusive if they cannot occur at the same time. In other words, the outcome of one of the events cannot affect the outcome of the other. Hence, tossing a coin once, twice, or even two rolls of a die are all independent events. For example, getting a head in the first toss cannot influence the outcome of the second toss (It can be a head or tail, each outcome with a probability of 0.5). However, drawing two cards from a deck is not independent, as the probability of the second draw will be affected by the result of the first.

FRM Part 1, Book 2: Quantitative Analysis, Chapter 1: Fundamentals of Probability, Learning objective: Describe independent events and mutually exclusive events

22. You have been given the following incomplete probability matrix about the return (in foreign currency terms) on an investment made in a foreign country as compared to the benchmark and the expected change in the exchange rate of the local currency against the foreign currency:

	Outperform	**Underperform**
Appreciation	10%	15%
No change		20%
Depreciation	25%	
Total		55%

What is the unconditional probability that the investment will outperform the market?

A. 45%
B. 40%
C. 55%
D. 50%

Correct answer: A

Explanation: The sum of all the probabilities in the column titled "outperform" will be the unconditional probability that the investment will outperform the market regardless of the change in the exchange rate. Similarly, the unconditional probability of underperformance will be the sum of the probabilities in the column titled "underperform." Since we know that this probability is 55%, the joint probability for "underperform" and "depreciation" can be found as follows:

55% = 15% + 20% + Joint probability of underperform and depreciation (X)

X = 20%

The sum of all joint probabilities in a matrix equal 100%. Therefore,

100% = 10% + 15% + Joint probability of outperform and no change (Y) + 20% + 25% + 20%

Y = 10%

Then, the unconditional probability that the investment will outperform the market = 10% + 10% + 25%

= 45%

	Outperform	**Underperform**
Appreciation	10%	15%
No change	10%	20%
Depreciation	25%	20%
Total	45%	55%

FRM Part 1, Book 2: Quantitative Analysis, Chapter 4: Multivariate Random Variables, Learning objective: Explain how a probability matrix can be used to express a probability mass function

23. A random variable Y has an equal probability of having a value of 0, 1, 2 or -1. Given that X = 3Y, what is the covariance of Y and X?

A. 3.25
B. 3.00
C. 3.50
D. 3.75

Correct answer: D

Explanation: The probability of Y being 0, 1,-1 or 2 is equal => ¼ for each value

The mean of Y can be calculated as:

$$E(Y) = \sum y_i P(Y = y_i) = ¼(0) + ¼(1) + ¼(-1) + ¼(2) = ½$$

$$E(X) = \sum x_i P(X = x_i) = ¼(0 \times 3) + ¼(1 \times 3) + ¼(-1 \times 3) + ¼(2 \times 3) = 3/2$$

The covariance between X and Y will be:

$$Cov(X,Y) = E(X - E[X])(Y - E[Y])$$

$$Cov(X,Y) = ¼(0 - ½)(0 - \frac{3}{2}) + ¼(1 - ½)(3 - \frac{3}{2}) + ¼(-1 - ½)(-3 - \frac{3}{2}) + ¼(2 - ½)(6 - \frac{3}{2})$$

$$Cov(X,Y) = \frac{3}{16} + \frac{3}{16} + \frac{27}{16} + \frac{27}{16}$$

$$Cov(X,Y) = 3.75$$

FRM Part 1, Book 2: Quantitative Analysis, Chapter 4: Multivariate Random Variables, Learning objective: Define covariance and explain what it measures.

24. All of the following statements are true, except:

A. If the return distributions of two investments have the same mean and standard deviation, the one which has more a negatively skewed distribution will be considered to be riskier
B. Distributions with positive excess kurtosis are known as leptokurtic
C. If two investments have the same mean, standard deviation and skewness, then the one with the lower kurtosis will be considered riskier
D. The normal distribution has a kurtosis of 3

Correct answer: C

Explanation: If two investments have similar return distributions in terms of mean and standard deviation and skewness, then the one with a higher kurtosis will be considered riskier as excess kurtosis implies more extreme points which results in more risk. (The possibility of experiencing extreme and financially devastating losses is higher).

FRM Part 1, Book 2: Quantitative Analysis, Chapter 5: Sample Moments, Learning objective: Estimate and interpret the skewness and kurtosis of a random variable.

25. Which of these probability distributions is most likely to be used to model stock prices?

A. Normal
B. Lognormal
C. Binomial
D. Poisson

Correct answer: B

Explanation: The normal and lognormal distributions are the most commonly used distributions for modeling the returns of investments.

However, a lognormal can be regarded as the exponential of a normal, so it cannot take negative values. This characteristic makes it suitable for dealing with stock prices, which also cannot take negative values.

FRM Part 1, Book 2: Quantitative Analysis, Chapter 3: Common Univariate Random Variables, Learning objective: Distinguish the key properties and identify the common occurrences of the following distributions: uniform distribution, Bernoulli distribution, binomial distribution, Poisson distribution, normal distribution, lognormal distribution, Chi-squared distribution, Student's t, and F-distributions

26. In 2016, DollarCom issued both BBB corporate bonds and common stocks. The following probability matrix summarizes the performance of the stock according to the ratings of the bonds:

		Stock Outperform	**Stock Underperform**
	Upgrade to BBB+	15%	5%
Bonds	**No change**	10%	20%
	Downgrade to BBB-	20%	30%

What is the probability that the credit ratings of the bonds are upgraded, given that the stock has underperformed?

A. 20%
B. 25%
C. 9%
D. 5%

Correct answer: C

Explanation: We need to find the probability P [Upgrade | Underperform]

P [Upgrade | Underperform] = $\frac{P\,[Upgrade \cap Underperform]}{P\,[Underperform]}$

Where the probability of underperformance is the sum of all probabilities in the column titled "Underperform.".

P [Upgrade | Underperform] = $\frac{5\%}{55\%}$

P [Upgrade | Underperform] = 9%

FRM Part 1, Book 2: Quantitative Analysis, Chapter 1: Fundamentals of Probability, Learning objective: Define and calculate a conditional probability

27. You have been given the following probabilities:

P [A | B] = 25%

P [B | A] = 50%

P [A] = 30%

What is P[B]?

A. 60%
B. 15%
C. 75%
D. 42%

Correct answer: A

Explanation: Use Bayes' theorem to find P [B].

$P\ [A \mid B] = \frac{P\ [B \mid A] \times P\ [A]}{P\ [B]}$

Therefore, $P\ [B] = \frac{P\ [B \mid A] \times P\ [A]}{P\ [A \mid B]} = \frac{0.5 \times 0.3}{0.25} = 0.6$

FRM Part 1, Book 2: Quantitative Analysis, Chapter 1: Fundamentals of Probability, Learning objective: Explain and apply Bayes' rule.

28. An analyst gathers monthly data about the returns of a stock for the past five years. If the mean monthly return is 6% and the standard deviation of the series of returns is 1.8%, then what is the standard deviation of the mean over the period?

A. 0.45%
B. 0.23%
C. 0.52%
D. 1.39%

Correct answer: B

Explanation: Standard deviation of the mean = $\frac{\sigma}{\sqrt{n}}$

$= \frac{1.8\%}{\sqrt{5 \times 12}}$

= 0.23%

FRM Part 1, Book 2: Quantitative Analysis, Chapter 6: Hypothesis Testing, Learning objective: Identify the steps to test a hypothesis about the difference between two population means.

29. An insurance policy reimburses dental expenses, X, up to a maximum benefit of 250. The probability density function for X is as follows:

$$f_X(x) = \begin{cases} ke^{-0.004x}, & x \geq 0 \\ 0, & elsewhere \end{cases}$$

Where k is a constant.

Calculate the median amount of the benefit.

A. 160
B. 172
C. 173
D. 250

Correct answer: C

Explanation: "We need to first find the value of k. We know that:

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

So that:

$$\int_{0}^{\infty} ke^{-0.004x}dx = 1$$

This implies that,

$$k\left[-\frac{1}{0.004}e^{-0.004x}\right]_0^{\infty} = 1$$

$$= k\left[0 - -\frac{1}{0.004}\right] = 1$$

Therefore, $k = 0.004$

$$P(X < m) = \int_{-\infty}^{m} f_X(x)dx = F(m) = 0.5$$

This implies,

$$\int_{0}^{m} 0.004e^{-0.004x}dx = 0.5$$

$$[-e^{-0.004x}]_0^m = 0.5$$

$$= [-e^{-0.004m} - -e^0] = 0.5$$

$$= -e^{-0.004m} + 1 = 0.5$$

$-e^{-0.004m} = -0.5$

Therefore,

$$m = \frac{\ln 0{,}5}{-0.004} \cong 173$$

FRM Part 1, Book 2: Quantitative Analysis, Chapter 2: Random Variables, Learning objective: Explain the effect of a linear transformation of a random variable on the mean, variance, standard deviation, skewness, kurtosis, median and interquartile range.

30. A zero-coupon bond with a notional value of $100 and price x has the following probability density function:

$$f(x) = \frac{x}{5000}\ for\ 0 \le x \le 100$$

Determine the probability that the price of the bond is between $80 and $90 inclusively.

A. 0.0017
B. 0.81
C. 0.17
D. 0.64

Correct answer: C

Explanation: Recall that for a continuous f(x) the probability P (a ≤ x ≤b) is given by:

$$P(a \le x \le b) = \int_a^b f(x)dx$$

In this case, $f(x) = \frac{x}{5000}$ and we need $P(80 \le x \le 90)$ So,

$$P(80 \le x \le 90) = \int_{80}^{90} \frac{x}{5000} dx$$

$$= \frac{1}{5000} \int_{80}^{90} x dx$$

$$= \frac{1}{5000} \left[\frac{x^2}{2}\right]_{80}^{90}$$

$$= 1/5000 \left[\frac{90^2}{2} - \frac{80^2}{2}\right]$$

$$= \frac{1700}{10000} = 0.17$$

FRM Part 1, Book 2: Quantitative Analysis, Chapter 2: Random Variables, Learning objective: Describe and distinguish a probability mass function from a cumulative distribution function and explain the relationship between these two.

31. For a variable v, the value of α is 2, and the probability that v > 100 is 0.1. What is the probability that v is greater than 200?

A. 0.025
B. 0.050
C. 0.105
D. 0.225

Correct answer: A

Explanation: According to the Power Law:

$P(v > x) = Kx^{-\alpha}$

Then,

$0.1 = K(100)^{-2}$

$K = 1000$

Then,

$P(v>200) = 1000\ x^{-2}$

$P(v>200) = 1000\ (200)^{-2}$

$P(v>200) = 0.025$

FRM Part 1, Book 2: Quantitative Analysis, Chapter 12: Measuring Returns, Volatility and Correlation, Learning objective: Describe the power law and its use for non-normal distributions

32. You have been provided the following set of values for an independent variable A and a dependent variable B:

A	B
1	5
2	8
3	12
4	15

What are the slope and the intercept of the regression line?

A. The slope is 3.4, and the intercept is 1.5
B. The slope is 2.6, and the intercept is 2.2

C. The slope is 3.1, and the intercept is 1.3
D. The slope is 2.8, and the intercept is 1.8

Correct answer: A

Explanation:

A	B	$A-\bar{A}$	$B-\bar{B}$	Cov (A,B)	Var(A)
1	5	-1.5	-5	7.5	2.25
2	8	-0.5	-2	1	0.25
3	12	0.5	2	1	0.25
4	15	1.5	5	7.5	2.25
Sum = 10	Sum = 40	0	0	17	5.00
Average = 2.5	Average = 10				

The slope is equal to the covariance of A and B divided by the variance of A.

Slope = 17/5 = 3.4

The intercept can be found by using the following equation:

$b_o = \bar{B} - b_1 \bar{A}$

$= 10 - 3.4\,(2.5)$

= 1.5

FRM Part 1, Book 2: Quantitative Analysis, Chapter 7: Linear Regression, Learning objective: Interpret the results of an ordinary least squares (OLS) regression with a single explanatory variable.

33. An ordinary least squares regression produces the following:

$$\sum(\boldsymbol{Y_i} - \bar{\boldsymbol{Y}})^2 = \mathbf{58}$$

$$\sum(\boldsymbol{Y_i} - \hat{\boldsymbol{Y}})^2 = \mathbf{10}$$

$$\sum(\hat{\boldsymbol{Y}} - \bar{\boldsymbol{Y}})^2 = \mathbf{48}$$

What is the coefficient of determination for the regression?

A. 0.792
B. 0.208
C. 0.828
D. 0.654

Correct answer: C

Explanation: The coefficient of determination or $R^2 = 1 - \frac{\sum(Y_i - \hat{Y})^2}{\sum(Y_i - \bar{Y})^2} = 1 - 10/58 = 0.828$

FRM Part 1, Book 2: Quantitative Analysis, Chapter 8: Regression with Multiple Explanatory Variables, Learning objective: Interpret goodness of fit measures for single and multiple regressions, including R^2 and adjusted-R^2.

34. The result of a two-variable regression analysis produces the following data:

	Coefficient
Constant	2.1
Slope	1.2

	Degrees of freedom	**Sum of squares**
Regression	1	35.4
Residual error	8	14.6

What is the correlation coefficient of the two variables?

A. 0.841
B. 0.708
C. 0.588
D. 0.767

Correct answer: A

Explanation: The coefficient of determination or $R^2 = 1 - \frac{SSR}{TSS} = 1 - \frac{14.6}{35.4 + 14.6} = 0.708$

The coefficient of correlation is the square root of the coefficient of determination = $\sqrt{0.708} = 0.841$.

FRM Part 1, Book 2: Quantitative Analysis, Chapter 8: Regression with Multiple Explanatory Variables, Learning objective: Interpret goodness of fit measures for single and multiple regressions, including R2 and adjusted-R2.

35. An analyst performs a regression of a stock's return on the overall market return using a sample of 2 years of monthly returns. He estimates the slope coefficient to be 0.55 with a standard deviation of 0.26. The analyst then performs a hypothesis test at a 5% level of significance to determine if the slope coefficient is significantly different from zero. Which of the following correctly describes the test statistic and result of this test?

A. The test statistic is 2.12, and the null hypothesis is rejected to conclude that the slope coefficient is different from zero
B. The test is 2.12, and the null hypothesis cannot be rejected to conclude that the slope coefficient is different from zero

C. The test is 0.47, and the null hypothesis cannot be rejected to conclude that the slope coefficient is different from zero
D. The test is 0.47, and the null hypothesis is rejected to conclude that the slope coefficient is different from zero

Correct answer: A

Explanation: A t-test will have to be performed to test the hypothesis.

$$T\ statistic = \frac{(b_1 - B_1)}{s_b} = \frac{0.55 - 0}{0.26} = 2.12$$

The critical two-tailed t-value with 22 degrees of freedom is 2.074. Since the t-value 2.12 is greater than the critical value of 2.07, we can reject the null hypothesis and conclude that the slope coefficient is different from 0.

FRM Part 1, Book 2: Quantitative Analysis, Chapter 7: Linear Regression, Learning objective: Explain the steps needed to perform a hypothesis test in linear regression.

36. If the variance of the residuals is not constant across all observations in a sample, the regression is said to be:

A. Leptokurtic
B. Multicollinear
C. Platykurtic
D. Heteroskedastic

Correct answer: D

Explanation: When the variance of the residuals is not constant across all observations in a sample, the regression is said to be heteroskedastic.

FRM Part 1, Book 2: Quantitative Analysis, Chapter 9: Regression Diagnostics, Learning objective: Explain how to test whether a regression is affected by heteroskedasticity.

37. Which of the following statements is false?

A. Multicollinearity occurs when two or more of the independent variables in a multiple regression are highly correlated with each other
B. Multicollinearity distorts the standard error of the regression and the coefficient of standard errors
C. If there exists imperfect multicollinearity, then it is not possible to find the ordinary least squares (OLS) estimators necessary for the regression results

D. As a result of multicollinearity, there is a greater probability that it will be incorrectly concluded that a variable is not statistically significant

Correct answer: C

Explanation: Multicollinearity occurs when two or more of the independent variables in a multiple regression are highly correlated with each other. The multicollinearity is said to perfect if one of the independent variables is a perfect linear combination of the other independent variables. In such a case, it is not possible to find the OLS estimators necessary for the regression results.

FRM Part 1, Book 2: Quantitative Analysis, Chapter 9: Regression Diagnostics, Learning objective: Characterize multicollinearity and its consequences; distinguish between multicollinearity and perfect collinearity.

38. An analyst runs a multiple regression of a variable Y on four independent variables for 50 observations. If the sum of squared errors for the regression is 125 and the total sum of squares is 450, then what is the adjusted R^2 of the regression?

A. 0.722
B. 0.786
C. 0.697
D. 0.643

Correct answer: C

Explanation: Coefficient of determination or $R^2 = 1 - \frac{SSR}{TSS} = 1 - 125/450 = 0.722$

Adjusted $R^2 = 1 - [\left(\frac{n-1}{n-k-1}\right) \times (1 - R^2)]$

Adjusted $R^2 = 1 - [\left(\frac{50-1}{50-4-1}\right) \times (1 - 0.722)]$

Adjusted $R^2 = 0.697$

The coefficient of correlation is the square root of the coefficient of determination = √0.708 = 0.841.

FRM Part 1, Book 2: Quantitative Analysis, Chapter 8: Regression with Multiple Explanatory Variables, Learning objective: Interpret goodness of fit measures for single and multiple regressions, including R^2 and adjusted-R^2

39. If the leading and lagged values of the mean, variance and covariance of a time series do not change with time, the time series is said to be:

A. Autocorellated
B. Autoregressive
C. Covariance stationary
D. Autocovariant

Correct answer: C

Explanation: If the leading and lagged values of the mean, variance and covariance of a time series do not change with time, it is said to be covariance stationary.

FRM Part 1, Book 2: Quantitative Analysis, Chapter 10: Stationary Time Series, Learning objectives:

- ***Define the autocovariance function and the autocorrelation function.***
- ***Define and describe the properties of autoregressive (AR) processes.***

40. Which of the following statements is (are) false?

I. An advantage of bootstrapping is that it allows to make inferences without making strong assumptions about the distribution

II. Another advantage of bootstrap is that it is not affected by outliers in the data

III. Bootstrapping will be ineffective if the data are not independent of one another

A. I only
B. II only
C. II & III only
D. All of the statements are false

Correct answer: B

Explanation: Bootstrapping is ineffective if there are outliers in the data, or if the data are not independent of one another.

FRM Part 1, Book 2: Quantitative Analysis, Chapter 13: Simulation Bootstrapping, Learning objective: Describe situations where the bootstrapping method is ineffective.

Financial Markets and Products

41. An unwanted result of deposit insurance is that it can cause banks to engage in risky behavior in the knowledge that depositors are well insured by the *Federal Deposit Insurance Corporation* (*FDIC*). This is an example of:

A. Adverse selection
B. Moral hazard
C. Asymmetric information
D. Systemic risk

Correct answer: B

Explanation: Moral hazard occurs when there is a lack of incentive to guard against risk as one is protected from its consequences. Deposits from customers form banks' main sources of funds. If a bank engages in excessively risky ventures, it is highly likely that depositors, being naturally risk-averse, will withdraw their funds. If however the deposits of a bank are insured by the government, the bank can take undue risks without fear of an adverse reaction from its depositors.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 1: Banks, Learning objective: Explain how deposit insurance gives rise to a moral hazard problem.

42. An insurance contract that lasts for a specified period and pays a lump sum either when the policyholder dies or at the end of the period, whichever comes first, is known as:

A. Term life insurance
B. Whole life insurance
C. Endowment life insurance
D. Universal life

Correct answer: C

Explanation: An insurance contract that lasts for a specified period and pays a lump sum either when the policyholder dies or at the end of the period is known as endowment life insurance.

A is incorrect. Under a term life policy, the benefit is payable if and only if the insured dies within the specified term, say 10 years.

B is incorrect. Under whole life insurance, there are no term limits, and the benefit is paid whenever the insured dies, subject to policy terms and conditions, including consistent premium payment.

D is incorrect. Universal life insurance is a form of whole life insurance where the insured is at liberty to reduce the premium payable down to a specified minimum without the policy lapsing.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 2: Insurance Companies and Pension Plans, Learning objective: Describe the key features of the various categories of insurance companies and identify the risks facing insurance companies.

43. An insurance company's financial statements contain the following information:

	(In million USD)
Premiums	120
Claims	65
Operating expenses	20
Investment income	5

The combined ratio for the insurance company is closest to:

A. 54%
B. 75%
C. 68%
D. 71%

Correct answer: D

Explanation: The combined ratio is the sum of the loss ratio and expense ratio of an insurance company.

$$Combined\ ratio = \frac{Claims + \ Operating\ expenses}{Premiums}$$

$$= \frac{65\ +\ 20}{120} = 71\%$$

FRM Part 1, Book 3: Financial Markets and Products, Chapter 2: Insurance Companies and Pension Plans, Learning objective: Calculate and interpret loss ratio, expense ratio, combined ratio, and operating ratio for a property-casualty insurance company.

44. An investor in a hedge fund requires a net return after fees of 18%. If the hedge fund charges fees of 1 plus 25%, how much will the hedge fund have to earn for the investor to get his desired return?

A. 22.30%
B. 25.00%
C. 25.70%
D. 24.00%

Correct answer: B

Explanation: The net return of the investor, R, can be expressed in terms of the hedge fund's return X as:

R = X – 0.01 – 0.25(X - 0.01)

Since the investor requires a return of 15%

0.18 = X – 0.01 – 0.25X + 0.0025

X = 25%

FRM Part 1, Book 3: Financial Markets and Products, Chapter 3: Fund Management, Learning objective: Calculate the return on a hedge fund investment and explain the incentive fee structure of a hedge fund, including the terms hurdle rate, high-water mark, and clawback.

45. Which of the following investment strategies does an equity market neutral fund follow?

A. Long-short equity
B. Dedicated short
C. Merger arbitrage
D. Convertible arbitrage

Correct answer: A

Explanation: A long-short equity strategy involves a manager taking opposing long and short positions in sets of stocks considered to be undervalued and overvalued by the market. An equity market neutral fund will follow this broad strategy by matching long and shorts in some way.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 3: Fund Management, Learning objective: Describe various hedge fund strategies, including long/short equity, dedicated short, distressed securities, merger arbitrage, convertible arbitrage, fixed income arbitrage, emerging markets, global macro and managed futures, and identify the risks faced by the hedge fund.

46. An investor buys a European put option on the shares of Yullu Corp. for a strike price of $25. Suppose the price of the share price falls to $20 at the time of the maturity of the option. If the cost of the put option was $2, then what is the net profit for the investor?

A. $20
B. $0
C. $3
D. $5

Correct answer: C

Explanation: A put option gives the buyer of the option the right to sell the underlying asset by a certain date for a certain price. The investor has the right to sell the shares of Yullu Corp. for a price of $25 on the expiration date of the contract (it being a European option). Since the price of the share fell to $20 at the time of maturity of the option, the investor will make a profit of $25 - $20 = $5 by selling the share for $25. Given that the price paid for the option was $2, the net profit will be $5 - $2 = $3.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 4: Introduction to Derivatives, Learning objective: Identify and calculate option and forward contract payoffs.

47. You have been given the following spot rates table:

Maturity	Spot Rate
0.5 year	4.0%
1 year	4.3%
1.5 years	4.9%
2 years	5.5%

If a 2-year bond with a principal of $1000 pays a semiannual coupon of 8% per year, then which of the following is closest to the theoretical price of the bond?

A. $1,046
B. $998
C. $1,002
D. $1,197

Correct answer: A

Explanation: The theoretical value of the bond can be found by finding the present value of the cash flows of the bond and by discounting them using the corresponding spot rates. A 2-year bond, which pays semi-annual coupons of 8%, will provide cash flows of $40 after 0.5, 1 and 1.5 years and a final cash flow of $1040 at maturity.

Then the value of the bond = $40e^{-0.04\times0.5} + 40e^{-0.043\times1} + 40e^{-0.049\times1.5} + 1040e^{-0.055\times2}$

Price = $1046.36

FRM Part 1, Book 3: Financial Markets and Products, Chapter 16: Properties of Interest Rates, Learning objective: Calculate the theoretical price of a bond using spot rates.

48. Two investors – A and B – invest $1000 each in two different financial instruments. Investor A receives a return of 5% with semiannual compounding while Investor B receives a return of 5% with quarterly compounding. After two years, what will be the terminal values of the two investments?

	Investor A	Investor B
A.	$1,103.8	$1,104.4
B.	$1,104.4	$1,103.8
C.	$1,215.5	$1,477.5
D.	$1,477.5	$1,215.5

Correct answer: A

Explanation: Terminal value of an investment = $A(1+\frac{R}{m})^{mn}$

Value of investment A = $1000(1+\frac{0.05}{2})^{2(2)}$ = $ 1,103.8

Value of investment B = $1000(1+\frac{0.05}{4})^{4(2)}$ = $ 1,104.4

FRM Part 1, Book 3: Financial Markets and Products, Chapter 16: Properties of Interest Rates, Learning objective: Calculate the value of an investment using different compounding frequencies

49. What is the duration of a 1.5-year 6% semiannual coupon bond with a face value of $100, if the yield on the bond is 10% per annum with continuous compounding?

A. 1.3918
B. 1.4009
C. 1.4553
D. 1.4949

Correct answer: C

Explanation:

Time	Cash Flow	Present Value	Weight	Time x Weight
0.5	3	$3 \times e^{-0.05} = 2.854$	3.029%	0.01515
1.0	3	$3 \times e^{-0.1} = 2.715$	2.881%	0.02881
1.5	103	$103 \times e^{-0.15} = 88.653$	94.090%	1.4114
Total		**94.222**	**1.000**	**1.4553**

FRM Part 1, Book 3: Financial Markets and Products, Chapter 16: Properties of Interest Rates, Learning objective: Calculate the duration, modified duration and dollar duration of a bond

50. An investor can immunize their portfolio against large parallel shifts in the zero curve by:

A. Choosing a portfolio of assets and liabilities with a net duration of 1, and net convexity of 0.
B. Choosing a portfolio of assets and liabilities with a net duration of 0, and net convexity of 1.
C. Choosing a portfolio of assets and liabilities with a net duration of 0, and net convexity of 0.
D. Choosing a portfolio of assets and liabilities with a net duration of 1, and net convexity of 1.

Correct answer: C

Explanation: A portfolio can be immunized against large parallel shifts in the yield curve by choosing the portfolio such that it has a net duration of zero and a net convexity of zero. Such a portfolio will, however, be exposed to nonparallel shifts of the yield curve.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 16: Properties of Interest Rates, Learning objective: Calculate the change in a bond's price given its duration, its convexity, and a change in interest rates.

51. An investor short sells 1,000 shares of XYZ Corporation in June when the price is $50 per share. In September, a dividend of $1.5 per share is paid by the company. The investor closes out the position in October when the price of the share is $45. Assuming there are no borrowing costs involved, what is the profit or loss for the investor?

A. $3,500
B. $6,500
C. $5,000
D. $1,500

Correct answer: A

Explanation: At the time of short selling in June, the investor borrows 1,000 shares of the company and sells them for $50 per share for a total inflow of 50 x 1,000 = $50,000. In September, the investor will pay $1.5 per share in dividend for a total outflow of 1.5 x 1,000 = $1,500. Finally, to close the short position the investor buys 1,000 shares at a price of $45 paying 45 x 1,000 = $45,000. Since there are no borrowing costs, the total profit (loss) is = 50,000 – 1,500 – 45,000 = $3,500.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 10: Pricing Financial Forwards and Futures, Learning objective: Define short-selling and calculate the net profit of a short sale of a dividend-paying stock.

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52. An asset that provides no income has a storage cost of $3 per unit, paid at the end of the year. If the spot price of the asset is $220 and the risk-free rate is 5%, the futures price for the asset is closest to:

A. $228.80
B. $238.11
C. $231.28
D. $234.28

Answer: D

Explanation: Futures price = $(S_o + U)e^{rT}$

Since U is the present value of the storage cost, it can be calculated as U = $3e^{-0.05\times 1}$ = 2.854

Futures price = $(220 + 2.854)e^{0.05\times 1}$

= $234.28

FRM Part 1, Book 3: Financial Markets and Products, Chapter 11: Commodity Forwards and Futures, Learning objectives:

- ***Describe the cost of carry model and illustrate the impact of storage costs and convenience yields on commodity forward prices and no-arbitrage bounds.***
- ***Compute the forward price of a commodity with storage costs.***

53. When the futures price is above the expected future spot price, the situation is known as

A. Backwardation
B. Contango
C. Cost of carry
D. None of the above

Correct answer: B

Explanation: When the futures price is below the expected future spot price, the situation is known as normal backwardation, and when the futures price is above the expected future spot price, the situation is known as contango.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 11: Commodity Forwards and Futures, Learning objective: Define and interpret normal backwardation and contango.

54. You have been given the following information regarding a stock index:

Current index level	$1,120
Risk-free rate	6% per annum
Dividend yield	2% per annum

What is the futures price for a six-month contract on the index?

A. $1,142.63
B. $1,165.70
C. $1,154.11
D. $1,137.60

Correct answer: A

Explanation: Futures price = $S_o e^{(r-q)T}$

$= 1120\ e^{(0.06-0.02)0.5}$

= $1,142.63

FRM Part 1, Book 3: Financial Markets and Products, Chapter 10. Pricing Financial Forwards and Futures, Learning objective: Calculate the value of a stock index futures contract and explain the concept of index arbitrage

55. You have been given the following details for two bonds:

	US Treasury Bond	Corporate Bond
Principal	$100	$100
Coupon	5%	6%

If coupons are paid on March 5 and September 5 of each year for both bonds, what is the interest accrued on both bonds between the period of March 5, 2017, and May 27, 2017?

A. $1.128 on the Treasury Bond and $1.353 on the Corporate Bond.
B. $1.128 on the Treasury Bond and $1.367 on the Corporate Bond.
C. $1.139 on the Treasury Bond and $1.353 on the Corporate Bond.
D. $1.139 on the Treasury Bond and $1.367 on the Corporate Bond.

Correct answer: B

Explanation: The interest accrued on a treasury bond is calculated based on the actual/actual convention, that is, actual number of days. There are 83 days between March 5 and May 27, 2017, while there are 184 days between the coupon payments of March 5 and September 5.

The interest accrued on the Treasury Bond will be 100 x 2.5% x (83/184) = $1.128.

The interest accrued on a corporate bond is calculated on a 30/360 convention. In other words, every month in a calendar year is assumed to have 30 days. On this basis, there are 82 days between March 5 and May 27, 2017 (= 25 days in March + 30 days in April + 27 days in May), while there are 180 days between the coupon payments of March 5 and September 5 (30 days every month for a total of six months).

Therefore, the interest accrued on the Corporate Bond will be 100 x 3.0% x (82/180) = $1.367.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 19: Interest Rate Futures, Learning objective: Differentiate between the clean and dirty price for a US Treasury bond; calculate the accrued interest and dirty price on a US Treasury bond.

56. The price of a Treasury bond, which has a coupon of 10%, is quoted as 103-05. If the current date is March 28, 2018, and the bond will mature on October 25, 2030, the cash price of the bond is closest to:

A. $107.43
B. $111.66
C. $103.46
D. $112.24

Correct answer: A

Explanation: Coupons are paid semiannually on government bonds, so coupon payments are made on April 25 and October 25. The last coupon was paid on October 25, 2017. In addition, Treasury bonds use the actual/actual convention. The number of days that have passed since the last payment is 154 (= 6 + 30 + 31 + 31 + 28 + 28) while the number of days between the two payment dates (reference dates) is 182. The cash price of the bond will be the quoted price and the interest accrued.

Quoted price = 103 05/32 or $103.15625

Interest accrued = 154/182 x 5% x 100 = $4.23

Cash price = $103.199 + $4.23 = $107.4290

FRM Part 1, Book 3: Financial Markets and Products, Chapter 19: Interest Rate Futures, Learning objective: Differentiate between the clean and dirty price for a US Treasury bond; calculate the accrued interest and dirty price on a US Treasury bond.

57. Which of the following bond is the cheapest to deliver if the Treasury bond futures price is 100-08?

	Price	Conversion Factor
A.	122-04	1.2001
B.	134-08	1.3190
C.	118-20	1.1568
D.	141-24	1.3987

Correct answer: D

Explanation: The cheapest-to-deliver bond is the bond with the lowest quoted price, calculated using the following formula:

Quoted Price – Futures Price x Conversion Factor

Bond A = 122.125 – 100.25 x 1.2001 = 1.815

Bond B = 134.250 – 100.25 x 1.3190 = 2.020

Bond C = 118.625 – 100.25 x 1.1568 = 2.656

Bond D = 141.750 – 100.25 x 1.3987 = 1.530

FRM Part 1, Book 3: Financial Markets and Products, Chapter 19: Interest Rate Futures, Learning objective: Describe the impact of the level and shape of the yield curve on the cheapest-to-deliver Treasury bond decision.

58. An increase in one of the following factors is likely to cause the price of an American call option to decrease. Which one?

A. Current stock price
B. Strike price
C. Volatility
D. Risk-free rate

Correct answer: B

Explanation: The price of an American call option will increase with an increase in the current stock price, time to expiration, volatility, and the risk-free rate, while it will decrease with an increase in the strike price and the amount of future dividends.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 13: Properties of Options, Learning objective: Identify the six factors that affect an option's price.

59. Which of the following correctly describes a protective put?

A. Long position in a put option and a short position in the underlying shares
B. Long position in a put option and a long position in the underlying shares
C. Short position in a put option and a long position in the underlying shares
D. Short position in a put option and short position in the underlying shares

Correct answer: B

Explanation: A protective put, also known as a married put, can be implemented by establishing a long position in a put option along with a long position in the underlying shares. A protective put helps investors to guard against the loss of unrealized gains. The maximum loss is limited to the purchase price of the underlying stock less the strike price of the put option and the premium paid.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 14: Trading Strategies, Learning objective: Explain the motivation to initiate a covered call or a protective put strategy.

60. A European call option on a non-dividend paying stock has the following details:

Current stock price	\$21
Strike price	\$20
Time to maturity	6 months
Risk-free interest rate	10%

What is the lower bound for the option price?

A. \$1.32
B. \$0
C. \$1.98
D. \$2.90

Correct answer: C

Explanation: Lower bound = $S_o - Ke^{-rT}$

$= 21 - 20e^{-0.10\times 0.5}$

= \$1.98

FRM Part 1, Book 3: Financial Markets and Products, Chapter 13: Properties of Options, Learning objective: Identify and compute upper and lower bounds for option prices on non-dividend and dividend paying stocks

61. You have been given the following information:

Price of European call option on stock X	\$1.5
Maturity of call option on stock X	6 months
Strike price of call option on stock X	\$42
Current price of a share of stock X	\$38
Expected dividend on stock X (to be paid two months from now)	\$0.75
Risk-free interest rate	6%

What is the price of a European put option that expires in 6 months and has a strike price of \$42?

A. \$5.00
B. \$4.50
C. \$3.76
D. \$4.00

Correct answer: A

Explanation: Put call parity can be expressed as $c + Ke^{-rT} + D = p + S_o$

Then the price of the European put is $p = c + Ke^{-rT} + D - S_o$

Price = $c + Ke^{-rT} + D - S_o$

$= 1.5 + 42e^{-0.06(0.5)} + (0.75e^{-0.06\left(\frac{2}{12}\right)}) - 38$

= $5.00

FRM Part 1, Book 3: Financial Markets and Products, Chapter 13: Properties of Options, Learning objective: Explain put-call parity and apply it to the valuation of European and American stock options, with dividends and without dividends and express it in terms of forward prices.

62. An increase in which of the following factors is likely to cause the price of a European put option to increase?

A. Current stock price, dividends and volatility
B. Strike price, dividends and volatility
C. Dividends, risk-free rate and volatility
D. Risk-free rate and volatility

Correct answer: B

Explanation: The price of a European put option will increase with an increase in the strike price, volatility, and amount of future dividends, while it will decrease with an increase in the current stock price and risk-free rate.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 13: Properties of Stock Options, Learning objective: Identify the six factors that affect an option's price.

63. A strategy where a long position in a call option is combined with a long position in a put option, such that both options have the same strike price and expiration date, is known as

A. Strangle
B. Bear spread
C. Straddle
D. Bearish option

Answer: C

Explanation: A straddle combines a long position in a call option with a long position in a put option with both options having the same strike price and expiration date, paying both premiums. As a result, the investor is able to make a profit regardless of whether the price of the security goes up or down, assuming there are significant changes in the market price of the underlying stock.

A is incorrect. A strangle is an options strategy where the investor holds a position in both a call and put with different strike prices but with the same maturity and underlying asset.

B is incorrect. A bear spread is an option strategy that seeks to maximize profit in the event that the price of the underlying security declines. It involves simultaneous purchase and sale of options (either puts or calls) where a higher strike price is purchased and a lower strike price sold.

D is incorrect. A bearish option is an option purchased by an investor who strongly believes that the price of the underlying will decline.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 14: Trading Strategies, Learning objectives:

- ***Describe the use and calculate the payoffs of various spread strategies.***
- ***Describe the use and explain the payoff functions of combination strategies.***

64. An investor wants to implement a strangle using put and call options on the shares of UUKL. Which of the following pairs of options is he most likely to choose?

A. Buy a call option with a strike price of \$30 expiring in 6 months and a buy a put option with a strike price of \$30 expiring in 6 months
B. Buy a call option with a strike price of \$30 expiring in 6 months and a buy a put option with a strike price of \$30 expiring in 3 months
C. Buy a call option with a strike price of \$30 expiring in 6 months and a buy a put option with a strike price of \$25 expiring in 6 months
D. Buy a call option with a strike price of \$30 expiring in 6 months and a buy a put option with a strike price of \$25 expiring in 3 months

Correct answer: C

Explanation: A strangle involves combining a long position in a call with a long position in a put. While both options should have the same expiration date, they should have different strike prices.

of the underlying will decline.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 14: Trading Strategies, Learning objectives:

- ***Describe the use and calculate the payoffs of various spread strategies.***
- ***Describe the use and explain the payoff functions of combination strategies.***

65. An investor who buys a put option with a strike price of \$25 and sells a put option with a strike price of \$20 is most likely following which of these strategies:

A. Bull spread
B. Box spread
C. Butterfly spread
D. Bear spread

Correct answer: D

Explanation: A bear spread strategy is implemented by buying a put option with one strike price and selling a put option with another strike price where the strike price of the option purchased is greater than the strike price of the option sold.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 14: Trading Strategies, Learning objective: Describe the use and calculate the payoffs of various spread strategies.

66. Suzy Wong, a finance student at the University of Tokyo, regularly invests her extra income in stocks and derivatives. She owns stocks of Safe Home Cleaning Inc., which is currently trading at $29. She believes the stock will trade below $29 if new regulations on cleaning companies are introduced next month. If she is interested in entering in an option position that gives her the right to sell her stocks at $29 if the price of the stock goes below $29, then suggest the most appropriate option position for Wong.

A. Long call option with the strike price of $29
B. Short call option with the strike price of $29
C. Long put option with the strike price of $29
D. Short put option with the strike price of $29

Correct answer: C

Explanation: Suzy Wong should purchase a put option or take a long position in a put option on the stocks of Safe Home Cleaning Inc. The put option should have the strike price of $29 (hypothetically), so if the price of the stock falls below the strike price of $29, Wong can exercise her right to sell her stocks at the price $29 per share.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 12: Options Markets, Learning objective: Describe the types, position variations, payoffs, and profits and typical underlying assets of options.

67. Which of the following statements about over-the-counter (OTC) derivatives is false?

A. OTC derivatives have flexible and negotiable terms
B. OTC derivatives have very good liquidity
C. The credit risk for OTC derivatives is bilateral
D. OTC derivatives have negotiable and non-standard maturities

Correct answer: B

Explanation: Over the counter (OTC) derivatives, are customized transactions between two parties. They have flexible and negotiable terms and non-standard maturities. Since the transactions are between two private parties, the credit risk is bilateral and there exists high liquidity risk.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 5: Exchanges and OTC Markets, Learning objective: Compare exchange-traded and OTC markets and describe their uses.

68. Which of these statements about exchange-traded derivatives is false?

A. Exchange-traded derivatives have standardized contract terms
B. Exchange-traded derivatives have low liquidity risk
C. Exchange-traded derivatives have high credit risk
D. Exchange-traded derivatives have standard maturities

Correct answer: C

Explanation: Exchange-traded derivatives are standardized contracts with the involvement of a central counterparty. They have standardized terms, including maturity dates. The involvement of a central counterparty ensures that these transactions have low credit and liquidity risk.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 5: Exchanges and OTC Markets, Learning objective: Identify the classes of derivative securities and explain the risk associated with them.

69. Which of these is most likely to be a disadvantage of having a central counterparty (CCP)?

A. Liquidity risk
B. Loss mutualisation
C. Lack of transparency
D. Adverse selection

Correct answer: D

Explanation: A CCP provides the advantage of transparency, offsetting, loss mutualisation, legal and operational efficiency, liquidity, and default management. The disadvantages of having a CCP include moral hazard, adverse selection, bifurcations, and procyclicality.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 6: Central Clearing, Learning objective: Describe advantages and disadvantages of central clearing of OTC derivatives.

70. The key risk for a Central Counterparty (CCP) is that of default of a clearing member. Such an event can also have certain knock-on effects. These knock-on effects can include all of the following, EXCEPT:

A. Failed auctions
B. Default of other clearing members
C. Resignations
D. Wrong-way risk

Correct answer: D

Explanation: The default of a member can lead to various knock-on effects. These can include default or distress of other clearing members, failed auctions, resignations, and reputational risk for the CCP. Wrong way risk, on the other hand, arises due to unfavorable dependencies, for example between the value of the margin held and the creditworthiness of clearing members.

FRM Part 1, Book 3: Financial Markets and Products, Chapter 6: Central Clearing, Learning objective: Identify and explain the types of risks faced by CCPs.

Valuation and Risk Models

71. Frank Ort is a junior trader at the options trading desk of an investment bank. He is reviewing a recent analysis of Tempo Construction LLC. Stocks of Tempo are currently trading at $47. According to the bank's analysts, the price of the stock will either go up or down by $8 in one year depending on the revenues that the company will generate. Since Ort is quite pessimistic on the future development of the company, he is thinking about opening a short position by buying put options on Tempo's stocks with a strike price of $45. The risk-free rate is assumed to be 2.1% per year.

What is the no-arbitrage price of the option that Ort is looking to buy?

A. $2.55
B. $2.57
C. $2.77
D. $2.97

Correct answer: B

Explanation: Let's denote Δ as the number of company shares that will make the position in the put option riskless.

Since the position is riskless, its value in both cases presented below should be the same.

$55 \times \Delta = 39 \times \Delta + 6 \Rightarrow \Delta = 0.375$

Riskless portfolios, in the absence of arbitrage opportunities, earn the risk-free rate of interest.

Value at T_0 = PV(Value at T1)

$47 \times 0.375 + p = 55 \times 0.375 \times e^{-2.1\%}$

$p = 2.57$

T0

# of Stocks	Δ
Stock Price	47
Option's Strike price	45
Option's Value	p
Value of Position	47*Δ + ?

T1

Case: Stock price increase	
# of Stocks	Δ
Stock	55
Option's Strike price	45
Option's Value	0
Value of Position	55*Δ

Case: Stock price decrease	
# of Stocks	Δ
Stock	39
Option's Strike price	45
Option's Value	6
Value of Position	39*Δ + 6

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 14: Binomial Trees, Learning objective: Calculate the value of an American and a European call or put option using a one-step and two-step binomial model.

72. An analyst at an options trading desk is valuing a 3-month European call option on the broad market value-oriented index. The index is currently trading at $14 and has a continuously compounded dividend rate of 1%. The analyst expects a price volatility of 12% per quarter. The risk-free rate is 5% per year, and the option has a strike price of $14.

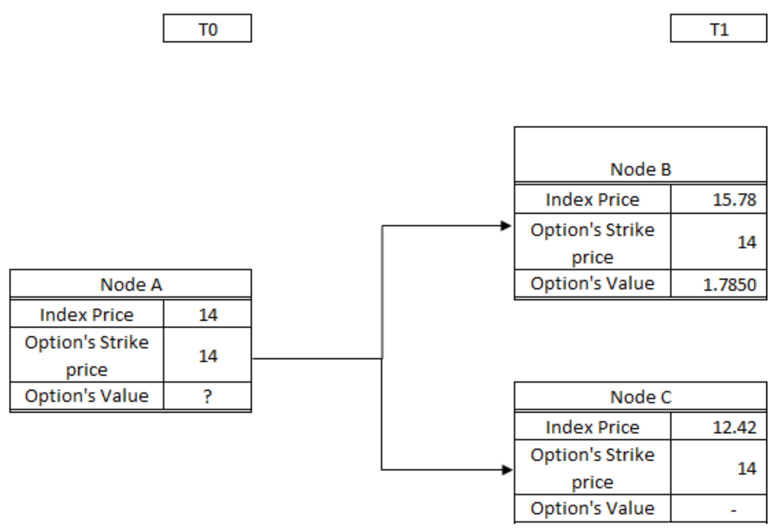
What is the price of the European call option?

A. $0.9022
B. $0.9045
C. $0.9207
D. $0.9232

Correct answer: A

Explanation: $p_{up} = \frac{e^{(r-r_{dividend})*\Delta t}-d}{u-d} = \frac{e^{(5\%-1\%)*0.25}-e^{-0.12}}{e^{0.12}-e^{-0.12}} = 0.5118$

The figure below shows the stock price and the corresponding option's intrinsic value at each node.



$\text{Price at Node A} = \left(1.785 * 0.5118 + 0 * (1 - 0.5118)\right) * e^{-5.0\% * \frac{3}{12}} = 0.9022$

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 14: Binomial Trees, Learning objectives:

- ***Calculate the value of an American and a European call or put option using a one-step and two-step binomial model.***
- ***Describe how volatility is captured in the binomial model***

73. Cytadela Bank from Poznan, Poland, needs to measure the yield of a coupon bond portfolio. One of the bonds in the portfolio has the maturity exactly equal to the term of a key rate, and the price of that bond is exactly par. How do the bond's yield and the key rate compare?

A. They are identical
B. The bond's yield is higher than the key rate
C. The bond's yield is lower than the key rate
D. The bond's yield has no discernible relationship with the key rate

Correct answer: A

Explanation: If the maturity of a coupon bond were exactly equal to the term of a key rate and if the price of that bond were exactly par, then that bond's yield would be identical to that key rate. By definition, then, that bond's key rate '01 with respect to that key rate would equal its yield-based DV01 while its key rate '01 with respect to all other key rates would be zero.

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 14: Modeling and Hedging Non-Parallel Term Structure Shifts, Learning objectives:

- ***Describe key-rate shift analysis.***
- ***Define, calculate, and interpret key rate '01 and key rate duration.***

74. You have been provided the following information:

Daily volatility estimate	2%
Decay factor	0.75
Most recent market return	3%

Using the Exponentially Weighted Moving Average Method (EWMA), what is the new estimate for volatility?

A. 2.101%
B. 2.783%
C. 2.291%
D. 1.876%

Correct answer: C

Explanation: Using the EWMA model:

$$\sigma^2 = \lambda {\sigma^2}_{n-1} + (1-\lambda){u^2}_{n-1}$$

$$\sigma^2 = 0.75\,(0.02)^2 + (1-0.75)(0.03)^2$$

$$\sigma^2 = 0.000525$$

$$\sigma = 2.291\%$$

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 3: Measuring and Monitoring Volatility, Learning objective: Apply the exponentially weighted moving average (EWMA) approach and the GARCH (1,1) model to estimate volatility

75. The current stock price of LLP Limited is \$22. The volatility is 20%, and the risk-free rate is 7%. According to the Black-Scholes-Merton model, what should be the price of a European call option on LLP's stocks with a strike price of \$20 and expiration of 3 months if $N(d_1)$ and $N(d_2)$ are assumed to be 0.5522 and 0.5027, respectively?

A. \$2.27
B. \$2.59
C. \$2.98
D. \$4.33

Correct answer: A

Explanation: The Black-Scholes formula is an expression for the current value of a European call option on a stock which pays no dividends before expiration of the option. The formula is:

$C = S \times N(d_1) - Xe^{-rT} \times N(d_2)$

Where:

S = current stock price

X = strike price

$= 22 \times 0.5522 - 20 \times \exp^{-7\% \times 0.25} \times 0.5027 = 2.27$

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 15: The Black-Scholes-Merton Model, Learning objective: Compute the value of a European option using the Black-Scholes-Merton model on a non-dividend-paying stock.

76. Five years ago, an investment bank provided a \$100,000,000 loan to RMS Constructions. The loan has the floating rate of 5% + 1-year Libor. The interest is paid annually, and the last interest payment was paid on the last day of 2016. Below is the information available for the 1-year Libor rates:

31/12/15	0.5%
31/12/16	2.1%
31/12/17 (forecast)	3.3%

What is the accrued interest as of 16/07/2017 if the bank and the client agreed to use the ACTUAL/360 day count convention?

A. $3,009,722.2
B. $3,885,277.8
C. $4,268,333.3
D. $4,541,944.4

Correct answer: B

Explanation: Actual number of days in period 31/12/16 - 16/07/17 = 197

Accrued coupon as of 16/07/17= $100,000,000 × (5% + 2.1%) × 197/360 = $3,885,277.80

Accrued interest as of the settlement date = 2.3125 × 132/182 = 1.6772

Clean price of the bond = 99.1808 - 1.6772 = 97.5035

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 9: Pricing Conventions, Discounting and Arbitrage, Learning objective: Differentiate between "clean" and "dirty" bond pricing and explain the implications of accrued interest with respect to bond pricing.

77. A fixed-income trader summarizes in the table below the prices of Treasury Bonds with semiannual coupon payment. The data is as of 01/01/17.

	Maturity	Coupon Rate	Price (per 100 face value)
Tranche 1	30/06/2017	3.5%	99-00
Tranche 2	31/12/2017	4%	100-16
Tranche 3	30/06/2018	5%	101-04

What are the discount factors for 0.5, 1, and 1.5 years?

A. d(0.5) = 0.9730; d(1) = 0.9662; d(1.5) = 0.9393
B. d(0.5) = 0.9551; d(1) = 0.9422; d(1.5) = 0.9102
C. d(0.5) = 0.9633; d(1) = 0.9523; d(1.5) = 0.9085
D. d(0.5) = 0.98990; d(1) = 0.9782; d(1.5) = 0.8787

Correct answer: A

Explanation: The cash flows are as follow:

	T(0.5)	T(1.0)	T(1.5)
Tranche 1	101.75	-	-
Tranche 2	2	102	-
Tranche 3	2.5	2.5	102.5

$99.00 = 101.75 \times d(0.5)$

$100.50 = 2 \times d(0.5) + 102 \times d(1)$

$101.125 = 2.5 \times d(0.5) + 2.5 \times d(1) + 102.5 \times d(1.5)$

d(0.5) = 0.9730

d(1) = 0.9662

d(1.5) = 0.9393

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 9: Pricing Conventions, Discounting and Arbitrage, Learning objective: Define discount factor and use a discount function to compute present and future values

78. An investment bank has a position in a callable bond. The risk management team of the bank prepared three different scenarios. The results are summarized in the table below:

Interest rate	Price of the callable bond	Price of the call option
8.70%	$101.750	$1.19
9.00%	$98.710	$1.01
9.30%	$95.988	$0.85

The CRO of the bank is evaluating the proposal of the structuring team to bifurcate the call option (e.g., to separate the call option from the bond and treat it as two different products: a non-callable bond and a call option) and sell it on the market. The CRO is primarily concerned with the impact of the bifurcation on the duration of the bond.

What is the effective duration (D) of the callable bond and a comparable non-callable bond?

A. D(callable bond) = 9.4384; D(non-callable bond) = 10.1986
B. D(callable bond) = 9.7288; D(non-callable bond) = 10.1986
C. D(callable bond) = 9.7288; D(non-callable bond) = 11.2873
D. D(callable bond) = 9.4384; D(non-callable bond) = 11.2873

Correct answer: B

Explanation: Recall that the that Effective duration is defined as:

$$D = -\frac{\Delta P}{P\Delta r}$$

Where

Δr = the size of a parallel shift in the interest rate term structure

ΔP – resultant change in the value of the position being considered

P = Value of the position being considered

Therefore, for the callable bond, when interest rate increases by 0.3%, the price decreases by $ 2.7220 (=95.988-98.710) and when the interest rate decreases by 0.3%, the price increases by $ 3.04 (=101.750-98.710). Therefore, the duration of the callable bond is given by:

$$= \frac{\frac{1}{2}(2.7220 + 3.04)}{98.710 \times 0.003} = 9.7288$$

Interest Rate	Price of the callable bond	Price of the call option	Price of the non-callable bond
8.70%	$101.750	$1.19	$102.940
9.00%	$98.710	$1.01	$99.720
9.30%	$95.988	$0.85	$96.838

For the non-callable bond, when interest rate increases by 0.3%, the price decreases by $ 2.8820 (=99.720-96.838) and when the interest rate decreases by 0.3%, the price increases by $ 3.22 (=102.940-99.720). Therefore, the duration of the callable bond is given by:

$$= \frac{\frac{1}{2}(2.8820 + 3.22)}{99.720 \times 0.003} = 10.1986$$

Note: We are approximating the change in price by averaging the price changes due to shifts in interest rates.

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 13: Applying Duration, Convexity, and DV01, Learning objective: Define, compute, and interpret the effective duration of a fixed income security given a change in yield and the resulting change in price.

79. A trader uses a model uses his valuation model to estimate the value of a bond portfolio at CAD 276.060 million. The term structure is flat. Using the same model, he estimates that the value of the portfolio would increase to CAD 278.935 million if all interest rates fell by 40 bps and would decrease to CAD 272.650 million if all interest rates rose by 40 bps. Using these estimates, determine the effective duration of the bond.

A. 2.85
B. 0.028
C. 0.023
D. 0.285

Correct answer: A

Explanation: Recall that the that effective duration is defined as:

$$D = -\frac{\Delta P}{P\Delta r}$$

When interest rate decreases by 0.4% (40 bps), the price increases by \$ 2.8750 (=99.720-96.838) and when the interest rate increases by 0.4%, the price decreases by \$ 3.41 (=276.060 -272.650). Therefore, the duration of the callable bond is given by:

$$= \frac{\frac{1}{2}(2.8750 \ + 3.41)}{276.060 \times 0.004} = 2.85$$

Note that this is similar to calculating duration at each change of interest (using the stated formula) rate then averaging the results. That is:

$$\begin{aligned} D = \frac{1}{2}(D^{-1} + D^{+}) &= \frac{1}{2}\left[\left(-\frac{2.8750}{76.060 \times -0.004}\right) + \left(-\frac{-3.41}{76.060 \times 0.004}\right)\right] \\ &= \frac{\frac{1}{2}(2.8750 \ + 3.41)}{276.060 \times 0.004} \\ &= 2.85 \end{aligned}$$

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 13: Applying Duration, Convexity, and DV01, Learning objective: Define, compute, and interpret the effective duration of a fixed income security given a change in yield and the resulting change in price.

80. John Wick, FRM, manages two assets – A and B. The correlation between A and B is 0.2. The table below outlines further details on the two assets:

Asset	Expected Return	Annual Volatility	Value
A	20%	20%	$200
B	30%	25%	$100

If Wick sells $100 worth of A and buys $100 worth of B, what would be the change in the daily VaR at the 99% level of confidence? Assume a normal distribution and 260 trading days.

A. 8.000
B. 0.897
C. 7.403
D. 0.102

The correct answer is B.

Explanation: First, compute the variance of the current portfolio. $Let\ W_A\ and\ W_B$represent the weight of A and B, respectively:

$$Variance_{current} = W_A^2 \times \sigma_A^2 + W_B^2 \times \sigma_B^2 + 2 \times W_A \times W_B \times \sigma_A \times \sigma_B \times Corr_{AB}$$

$$= 200^2 \times 0.2^2 + 100^2 \times 0.25^2 + 2 \times 200 \times 100 \times 0.2 \times 0.25 \times 0.2 = 2{,}625$$

$$VaR_{current} = \sqrt{2{,}625} \times 2.33 \times \frac{1}{\sqrt{260}} = 7.403$$

The new portfolio has positions of $100 in A and $200 in B.

$$Variance_{new} = 100^2 \times 0.2^2 + 200^2 \times 0.25^2 + 2 \times 100 \times 200 \times 0.2 \times 0.25 \times 0.2 = 3{,}300$$

$$VaR_{new} = \sqrt{3{,}300} \times 2.33 \times \frac{1}{\sqrt{260}} = 8.300$$

$$VaR_{new} - VaR_{current} = 8.3 - 7.403 = 0.897$$

The new VaR is higher because of the greater weight on asset B, which has higher volatility. Note that the expected returns are irrelevant.

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 2: Calculating and Applying VaR, Learning objective:Describe and calculate VaR for linear derivatives.

81. A trader wants to price a bond with a 5% coupon rate and a maturity of 1.5 years. The coupon payment is semiannual. Due to a bug in the system, the trader was not able to download the complete structure of the market spot rates.

The table below shows the data the trader was able to gather.

Term in years	Spot rates
0.5	1.10%
1.0	n/a
1.5	3.00%

Which of the following are the missing spot rate and the bond price if the 1-year discount factor is 0.9831?

A. 1-year spot rate = 1.71%; Bond price = 102.967
B. 1-year spot rate = 2.10%; Bond price = 102.957
C. 1-year spot rate = 2.50%; Bond price = 102.947
D. 1-year spot rate = 2.65%; Bond price = 104.944

Correct answer: A

Explanation: 1-year spot rate = $((1/0.9831)^{0.5} - 1) \times 2 = 1.71\%$

Bond Price = $2.5/(1+(1.10\%)/2)^{(0.5\times2)} + 2.5/(1+(1.71\%)/2)^{(1\times2)} + 102.5/(1+(3.00\%)/2)^{(1.5\times2)} = 102.967$

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 10: Interest Rates, Learning objective: Interpret the relationship between spot, forward, and par rates.

82. An analyst at a bond trading desk wants to decompose the P&L of a $5,000,000 face value U.S. Treasury 7 1/2s of April 31, 2030, over a six-month holding period. Currently, the bond trades at par. The position is financed with direct repo which has an interest rate of 3%.

If the market experiences a +100 basis-point parallel shift in the yield curve at the end of the sixth month, what is the cash-carry amount over the six month period?

A. -$500,000
B. -$75,000
C. +$112,500
D. +$187,500

Correct answer: C

Explanation: Cash-Carry = Coupon income – Financing cost = $5,000,000 × (7.5% - 3%) / 2 = $112,500

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 11: Bond Yields and Return Calculations, Learning objective: Explain the decomposition of the profit and loss (P&L) for a bond position or portfolio into separate factors including carry roll-down, rate change and spread change effects.

83. Amelia Stone plans to retire next year and is considering an investment that would help to ensure her future financial stability. On average, Stone spends $70,000 annually. After retirement, she plans to relocate to the countryside, a move that should cut her living expenses by 20%. She was approached by her broker with the idea to start investing in perpetuities. Specifically, he suggested a perpetuity that would yield 7% per year.

Assuming the mentioned reduction of costs, how much should Stone invest in the perpetuities to meet her new spending needs?

A. $800,000
B. $8,000,000
C. $560,000
D. $5,600,000

Correct answer: A

Explanation: New spending needs = $70,000 × 0.8 = $56,000

PV of perpetuity = C/I = $56,000/7% = $800,000

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 11: Bond Yields and Return Calculations, Learning objective: Calculate the price of an annuity and a perpetuity.

84. A risk manager in an investment bank was asked to forecast the sovereign rating of countries where the bank has significant holdings. The risk manager found sovereign transition rates across the Major Rating Categories 1995-2008 published by Fitch.

Annual Rating Transitions (%, Average Annual)

	AAA	AA	A	BBB	BB	B	CCC to C	D	Total
AAA	99.42	0.58	-	-	-	-	-	-	100.00
AA	4.12	94.12	1.18	-	-	0.59	-	-	100.00
A	-	3.55	92.91	3.55	-	-	-	-	100.00
BBB	-	-	8.11	87.84	3.38	0.68	-	-	100.00
BB	-	-	-	9.04	83.51	5.85	-	1.60	100.00
B	-	-	-	-	12.12	84.09	3.03	0.76	100.00
CCC to C	-	-	-	-	-	23.08	53.85	23.08	100.00

Given the table above, which of the below statements is correct?

A. A country has a 98.13% chance to have a rating of AA.
B. A country with a rating of BB has a 5.85% chance of a downgrade in one year.
C. AAA-rated countries will have the same rating in the future with a 99.42% probability.
D. There is a 7.97% probability that a country with a rating of AA will have an improvement of rating to AAA in two years.

Correct answer: D

Explanation: A country with a rating AA will become AAA in two years if it either has:

- an improvement of rating to AAA in year 1 (4.12% probability) and has the same AAA rating in year 2 (99.42% probability); or
- the same AA rating in year 1 (94.12% probability) and has an improvement of rating in year 2 (4.12% probability).

Therefore, the probability of an AAA rating in 2 years = $0.0412 \times 0.9942 + 0.9412 \times 0.0412 = 0.0797$

Option A is incorrect. The annual rating transitions table presents only data on annual transitions of rating.

Option B is incorrect. A country with a rating of BB has a 7.45% (5.85%+1.60%) chance of a downgrade in one year.

Option C is incorrect. AAA-rated countries will have the same rating in one year with a 99.42% probability.

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 4: External and Internal Rating, Learning objective: Describe external rating scales, the rating process, and the link between ratings and default.

85. For which of the following reasons are rating agencies most usually criticized?

A. Being too optimistic in the assessment of corporate ratings and too pessimistic in the assessment of sovereign ratings
B. Failure to align ratings with each other before publishing (e.g., one rating agency may downgrade a sovereign rating of the country to speculate grade, while the other can still keep it with investment grade)
C. Taking too long to change a rating
D. Charging high fees for publishing corporate and sovereign ratings

Correct answer: C

Explanation: It has long been argued that rating agencies take too long to change ratings and that these changes happen too late to protect investors from a crisis.

Option A is incorrect. Rating agencies have been accused of being too optimistic in their assessments of both corporate and sovereign ratings.

Option B is incorrect. When one rating agency lowers or raises a sovereign rating, other rating agencies seem to follow suit. This herd behavior reduces the value of having three separate rating agencies since their assessments of sovereign risk are no longer independent.

Option D is incorrect. Rating agencies usually offer sovereign ratings free of charge.

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 4: External and Internal Rating, Learning objective: Explain historical failures and potential challenges to the use of credit ratings in making investment decisions

86. A bank has the following internal rating transition matrix:

Annual Rating Transitions (%, Average Annual)

	A	**B**	**C**	**D**
A	98.50	1.00	0.50	-
B	1.00	88.00	7.50	3.50
C	-	10.00	75.50	14.50

Based on the matrix, what is the probability of default over 2 years of a company with a rating of B?

A. 2.5%.
B. 4.6%.
C. 6.8%.
D. 7.7%.

Correct answer: D

Explanation: A company with a rating of B will default in two years if it either:

- it defaults in Year 1 (3.5% probability); or
- has the same rating of B in year 1 (88.0% probability) and defaults in Year 2 (3.5% probability); or
- experiences a rating drop to C in year 1 (7.5% probability) and defaults in Year 2 (14.5% probability).

Therefore, the probability of the default over 2 years $= 0.035 + 0.88 \times 0.035 + 0.075 \times 0.145 = 0.077$.

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 4: External and Internal Ratings, Learning objective: Describe and interpret a rating transition matrix and explain its uses.

87. An intern in a risk management department of a bank is struggling with the understanding of the nature of credit ratings provided by the rating agencies. He overhears a discussion from some of his colleagues regarding the recent downgrade of company XYZ by S&P from A to BBB.

What is the interpretation of XYZ's credit rating downgrade from A to BBB?

A. Bonds issued by XYZ are currently overvalued in the market
B. Stocks of XYZ are expected to underperform the market benchmark in the nearest future
C. XYZ's capacity to meet its obligations slightly deteriorated from 'extremely strong' to 'very strong'
D. XYZ still has an adequate capacity to meet its obligations, but now it is more susceptible to a negative change in the business environment

Correct answer: D

Explanation: The credit rating of A reflects a company's strong capacity to make payments.

Credit ratings of BBB reflect a company's adequate payment capacity, although a negative change in the business environment could affect its capacity for repayment.

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 4: External and Internal Ratings, Learning objective: Describe external rating scales, the rating process and the link between ratings and default.

88. What is the role of the internal audit in stress testing governance?

A. Establishing adequate stress testing policies and procedures and ensure compliance with them
B. Monthly statistical back-testing of the stress test estimates against realized outcomes
C. Provision of an independent evaluation of the ongoing performance, integrity, and reliability of stress-testing activities
D. Appropriate resource allocation to ensure proper functioning of stress-test processes

Correct answer: C

Explanation: The internal audit should provide independent evaluation of the ongoing performance, integrity, and reliability of stress-testing activities. In other words, the audit department acts as a checker, hence ensuring adherence to stipulated policies and risk management procedures.

Options A and D are incorrect. These are the responsibilities of senior management.

Option B is incorrect. Stress tests, by definition, aim to estimate the potential impact of rare events. Statistical backtesting of stress-test estimates against realized outcomes may not be feasible because of the lack of the data.

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 8: Stress Testing, Learning objective: Describe the important role of the internal audit in stress testing governance and control.

89. The new CRO of a regional bank is trying to implement a new stress testing process for the bank. He asks one of the colleagues to start the preparation of stress testing policies, procedures and documentations. The CRO makes the following statements:

I. The policies should be transparent to third parties to ensure an understanding of the bank's stress testing activities.
II. The policies should describe the frequency and priority with which stress-testing activities should be conducted.
III. The policies should be reviewed and updated as necessary to ensure that stress testing practices remain up to date with changes in market conditions and the bank’s strategies.
IV. The policies should describe the overall purpose of stress-testing activities.

Which of the statements is/are correct?

A. Statement I and
B. Statement II and III
C. Statement II, III and IV
D. All of the statements are correct

Correct answer: D

Explanation: All of the statements are correct.

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 8: Stress Testing, Learning objective: Identify elements of clear and comprehensive policies, procedures and documentation for stress testing

90. An investment bank has $1,000,000,000 portfolio of senior unsecured loans evenly distributed between 10 different clients, each with an internal rating B. The CRO of the bank wants to stress the loan portfolio under a scenario of an economic slowdown with the GDP dropping by 2% and the unemployment increasing by 1%. Based on the prior experience, the CRO knows the economic slowdown scenario will negatively impact the credit quality of the borrowers, and lead to a rating deterioration from B to C for half of the clients. The recovery rates of senior unsecured facilities are assumed to be 60% in normal conditions and 40% in stress conditions.

Credit Rating	**PD**
A	0%
B	3%
C	25%
D	100%

What is the bank's loan portfolio expected loss in both the base and the stressed scenarios?

A. Loss (*Base scenario*) = \$12,000,000; Loss (*Stressed scenario*) = \$56,000,000
B. Loss (*Base scenario*) = \$12,000,000; Loss (*Stressed scenario*) = \$84,000,000.
C. Loss (*Base scenario*) = \$18,000,000; Loss (*Stressed scenario*) = \$56,000,000.
D. Loss (*Base scenario*) = \$18,000,000; Loss (*Stressed scenario*) = \$84,000,000.

Correct answer: B

Explanation: Expected loss (Base scenario) = 3% × 40% × \$1,000,000,000 = \$12,000,000

Expected loss (Stressed scenario) = 25% × 60% × \$500,000,000 + 3% × 60% × \$500,000,000 = \$84,000,000

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 8: Stress Testing, Learning objective: Identify key aspects of stress testing governance, including choice of scenarios, regulatory specifications, model building, stress-testing coverage, capital, and liquidity stress testing, and reverse stress testing.

91. You have been provided the following information:

Weighted long-run variance	0.000010
Weighting on previous period's return	0.15
Weighting on previous volatility estimate	0.80
Daily volatility estimate	1.5%
Recent market return	1%

Using the GARCH(1,1) model, the volatility per day on day n (σ_n) is closest to:

A. 1.109%
B. 1.276%
C. 1.432%
D. 1.231%

Correct answer: C

Explanation: Variance per day of a market variable on day n:

$$\sigma^2{}_n = \omega + \beta\sigma^2{}_{n-1} + \alpha u^2{}_{n-1}$$

$$\sigma^2{}_n = 0.00001 + 0.80(0.015)^2 + 0.15(0.01)^2$$

$$\sigma^2{}_n = 0.000205$$

Variance per day of a market variable on day n:

$\sigma_n = 1.432\%$

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 3: Measuring and Monitoring Volatility, Learning objective: Apply the exponentially weighted moving average (EWMA) approach and the GARCH (1,1) model to estimate volatility

92. Risk managers of a regional bank are discussing the advantages of stress testing. They make the following statements:

I. Based solely on historical data, stress testing provides forward-looking assessments of risk.
II. Stress testing plays an important role in capital and liquidity planning procedures.
III. Stress testing is facilitating the development of risk mitigation or contingency plans across a range of stressed conditions

Which of these statements is/are correct?

A. Statement I.
B. Statements I and II.
C. Statements I and III.
D. Statements II and III.

Correct answer: D

Explanation: Statement I is incorrect as stress testing scenarios should not be necessarily based on historical data. Current data and simulated but possible outcomes can also be used.

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 8: Stress Testing, Learning objective: Identify the advantages and disadvantages of stressed risk metrics

93. Which of the following statements regarding recommendations to supervisors is incorrect?

A. Supervisors should evaluate whether the scenarios are consistent with the risk appetite the bank has set for itself
B. Supervisors should take corrective actions if material deficiencies in the stress testing program are identified or if the results of stress tests are not adequately taken into consideration in the decision-making process
C. Supervisors should verify the active involvement of senior management in the stress testing program and require a bank to submit at regular intervals the results of its firm-wide stress testing program
D. Supervisors should verify that stress testing forms an integral part of the ICAAP and the bank's liquidity risk management framework

Correct answer: B

Explanation: Supervisors should not take corrective action on their own. Instead, they should require the management of the company to take corrective actions. To do this, supervisors should maintain open lines of communication with senior management.

All other statements are correct.

***FRM Part 1, Book 4: Valuation and Risk Models, Chapter 8: Stress Testing,* Learning objective: Describe the responsibilities of the board of directors and senior management in stress testing activities.**

94. In the run-up to the 2007/2008 financial crisis, Trevor Smith inherited $10,000,000 from his grandfather. During the crisis, Smith suffered huge losses in his equity holdings, and he now wants to invest the inheritance with minimal risks. He is considering the following options:

	Product	Interest rate	Interest rate compounding
Option 1	Deposit at an AA-rated bank	3.0500%	monthly
Option 2	Deposit at an AA-rated bank	3.1000%	semiannually
Option 3	Deposit at an AA-rated bank	3.1000%	annually
Option 4	Deposit at an AAA-rated bank	3.0500%	continuously

Which of these options has the highest monthly compounded interest rate?

A. Option 1 has the highest monthly compounded rate of 3.0500%.
B. Option 2 has the highest monthly compounded rate of 3.0802%.
C. Option 3 has the highest monthly compounded rate of 3.0868%.
D. Option 4 has the highest monthly compounded rate of 3.0539%.

Correct answer: B

Explanation:

	Interest rate compounded monthly	Comment
Option 1	3.0500%	-
Option 2	3.0802%	$= ((1 + 3.10\% / 2)^{1/6} - 1) \times 12$
Option 3	3.0568%	$= ((1 + 3.10\%)^{1/12} - 1) \times 12$
Option 4	3.0539%	$= (\exp(3.05\%/12) - 1) \times 12$

Therefore, option 2 has the highest return.

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 10: Interest Rates, Learning objective: Calculate and interpret the impact of different compounding frequencies on a bond's value.

95. The stock price of AlphaFarm skyrocketed to $200 after a recent announcement of a drug approval by the FDA. Francesca Merc wants to quickly estimate the approximate price of her European call option on Alpha's stocks using the Black-Scholes-Merton model. The call option has expiry in 3 months and a strike price of $50.

If the risk-free rate is 8%, then which of the following is closest to the price of Merc's call option?

A. $151
B. $158
C. $147
D. $140

Correct answer: A

Explanation: In cases when call options are significantly in the money:

$N(d_1)$ and $N(d_2)$ => 1

$c = S_0 \times N(d_1) - K \times e^{-t \times T} \times N(d_2) \approx S_0 - K \times e^{-r \times T} \approx 200 - 50 \times e^{-8\% \times 0.25} = 151$

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 15: The Black-Scholes-Merton Model, Learning objective: Compute the value of a European option using the Black-Scholes-Merton model on a non-dividend-paying stock.

96. If the covariance between Canadian and American interest rates is 0.092, and the variances of interest rates in Canada and the U.S. are 12.32% and 11.04%, respectively, then which of the following is closest to the correlation between Canadian and American interest rates?

A. 0.7889
B. 9.7641
C. 0.1478
D. 1.2677

Correct answer: A

Explanation: Correlation = Covariance / (Standard deviation (A) × Standard deviation (B))
Correlation = 0.092 / ($\sqrt{0.1232} \times \sqrt{0.1104}$) = 0.7889

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 2: Calculating and Applying VaR, Learning objective: Describe and calculate VaR for linear derivatives.

97. Consider a $75 call option on a non-dividend paying stock where the stock price is $68.30, the risk-free rate is 5%, the time to maturity is 120 days and N' = 0.3311. A 1% increase in the volatility will increase the value of the option by approximately:

A. 0.1424
B. 0.0743
C. 0.1297
D. 2.4873

Correct answer: C

Explanation: The Vega of a call option = $S_0 \times \sqrt{T} \times N'(d1)$ = $68.30 × $\sqrt{(120/365)}$ × 0.3311 = 12.97

Thus, a 1% (0.01) increase in the volatility increases the value of the option by approximately 0.01 × 12.97 = 0.1297

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 16: Option Sensitivity Measures: The Greek Letters, Learning objective: Define and describe theta, gamma, vega, and rho for options positions and calculate the gamma and vega for a portfolio.

98. Arab Bank has taken credit exposure to two corporate clients. The credit risk characteristics of these two loans have been provided to you:

Loan to BMG Electrics

Sanctioned amount	USD 730 million
Exposure amount	USD 630 million
Probability of default over the next one year	2.1%
Loss rate if the customer defaults	24%
Standard deviation of the probability of default	5%
Standard deviation of the loss rate	38%

Loan to FFA Construction

Sanctioned amount	USD 525 million
Exposure amount	USD 300 million
Probability of default over the next one year	1%
Loss rate if the customer defaults	25%
Standard deviation of the probability of default	3%
Standard deviation of the loss rate	25%

The correlation between the two loan accounts is 0.45.

What is the unexpected loss of the loan portfolio held by Nicolson Finance?

A. USD 48.86 million
B. USD 38.03 million
C. USD 39.65 million
D. USD 17.13 million

Correct answer: C

Explanation: Unexpected loss = Exposure amount × √{Probability of default × (Standard deviation of loss rate)2 + Loss rate2 × Standard deviation of probability of default2}

Unexpected loss (BGM) = USD 630 million × √{0.021 × (0.38)2 + 0.242 × 0.052} = USD 35.50 million

Unexpected loss (FFA) = USD 300 million × √{0.01 × (0.25) 2 + 0.252 × 0.032} = USD 7.83 million

Unexpected loss on the portfolio = √{Unexpected loss BGM2 + Unexpected loss FFA2 + (2 ×

Unexpected loss GBM × Unexpected loss FFA × correlation)}

Unexpected loss on the portfolio = √{35.52 + 7.832 + (2 × 35.5 × 7.83 × 0.45)} = USD 39.65 million

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 6. Measuring Credit Risk, Learning objective: Define and explain unexpected loss (UL).

99. Bakster Bank is following the basic indicator approach for calculating the operational risk amount for the year 2016. Some past financial details of the bank are given below:

Year	Net interest income	Non-interest income
2015	102	25
2014	104	22
2013	98	21
2012	91	17

(All $ amounts are in USD millions)

Based on the original Basel Accord, the bank must hold capital for operational risk for 2016 equal to:

A. USD 18 million
B. USD 18.6 million
C. USD 24.8 million
D. USD 31 million

Correct answer: B

Explanation: Based *on* the original Basel Accord, banks using the *basic indicator approach* must hold *capital* for *operational risk equal* to the average *over* the *previous three years* of a fixed percentage of positive *annual gross income*. Gross income is defined as net interest income plus non-interest income.

Net interest income plus non-interest income for the last three previous years:

Year	Net interest income	Non-interest income	Gross income
2015	102	25	127
2014	104	22	126
2013	98	21	119

Average gross income over the previous three years = (127 + 126 + 119)/3 = 124

Operational risk = 15% of average gross income over the previous three years = USD 124 million × 0.15 = USD 18.6 million

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 7: Operational Risk, Learning objective: Compare the basic indicator approach, the standardized approach and the advanced measurement approach for calculating operational risk regulatory capital.

100. Elisabeth Yung is the senior fixed income investment manager at a regional bank. She is interested in the investment in Rocks Corporation's medium-term (7-year) zero-coupon bond. The bond has a yield to maturity of 4.5% compounded annually and a daily yield volatility of 0.59%. What are the daily 99% VaR and the Macaulay duration of the bond if Yung decides to buy the nominal value of $10,000,000?

A. Daily 99% VaR = $920,851.67; Macaulay duration = 7
B. Daily 99% VaR = $676,668.02; Macaulay duration = 7
C. Daily 99% VaR = $676,668.02; Macaulay duration = 6.7
D. Daily 99% VaR = $920,851.67; Macaulay duration = 6.7

Correct answer: B

Explanation: The key point here is that Macaulay duration of zero coupon bond is equal to the maturity (7).

In the calculations of VaR, we need to use modified duration.

$$\text{VaR of the position } = \frac{0.59\% \times 2.33 \times 7}{1 + 4.5\%} \times \frac{\$10{,}000{,}000}{(1 + 4.5\%)^7} = \$676{,}668.02$$

FRM Part 1, Book 4: Valuation and Risk Models, Chapter 2. Calculating and Applying VaR, Learning objective: Describe and calculate VaR for linear derivatives.